WELL-POSED SELF-SIMILARITY IN INCOMPRESSIBLE FLOWS UNDER STANDARD CONDITIONS

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Abstract. This article reviews the properties of the self-similar solutions of the Navier-Stokes equation for incompressible fluids. Since any smooth solution can be embedded into a self-similar solution at the identity scale, it follows that in the standard parabolic system, the initial solution will remain smooth for all time as long as the self-similar solution is selected to have certain isobaric weight. The method was also applied to the incompressible equation of Euler, resulting in smooth solutions for all time.

1. Introduction

The Navier-Stokes equations (NSE) lie at the heart of fluid dynamics and proving their regularity is among the most challenging problems in modern mathematics [1]. The NSE are given by

$$\begin{aligned} \rho\left(\frac{\partial \vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u}\right) &= -\nabla p + \mu\Delta\vec{u} \\ \nabla\cdot\vec{u} &= 0, \end{aligned} \tag{1.1}$$

where $\vec{u}(x, y, z, t) = (u, v, w)$ represents the fluid velocity, p the pressure, and $\nu = \mu/\rho$ the kinematic viscosity. For the sake of simplicity, no external forces are acting on the fluid.

In this and previous works [2], we build on Bouton's invariant theory [3], which apply Lie's theory of continuous groups to scaling transformations. This method can be applied to any ordinary or partial differential equation that exhibits scaling invariance. The technique has been employed very eloquently for example, by Lloyd [4], although stopped short from fully exploiting the method's potential; the derived NSE invariants were considered to have exhausted the abilities of the method.

However, the method of Bouton-Lie can be more fully utilized to derive all self-similar NSE solutions and establish their global regularity [2]. Although, the analysis remains limited to that specific subclass. What still needs to be considered are solutions that arise from non-self-similar initial data. Especially, smooth solutions arising from Schwartz-class or space-periodic initial data are of particular interest and need to be studied in detail [1].

For these reasons, in the present article we extend the analysis of Ref. [2], demonstrating that the self-similar solutions can arise from both smooth, space-periodic initial data and Schwartz-class initial data. Under the standard parabolic conditions, certain self-similar solutions preserve the boundedness of the system's energy under arbitrary rescaling, thereby avoiding the uncontrolled rescaling inherent in the classical Navier–Stokes equations that could otherwise act as a blowup mechanism.

The potential of self-similar structures is recognized and well-established in NSE analysis [5], [6]. E.g., Giga and Miyakawa constructed forward self-similar solutions in Morrey spaces, showing that such forms can arise naturally and govern the long-time behavior of the flow under smallness assumptions. The present work extends this perspective: by embedding arbitrary smooth, divergence-free initial data into globally defined self-similar families, we show that the protective effect of self-similarity can be applied beyond small-data regimes, yielding globally smooth solutions consistent with the Clay regularity framework [1].

Our analysis begins with getting the reader acquainted with the basic facts and theorems of invariant theory, followed by an outline of the derivation of the self-similar solutions of the NSE. We then focus

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on a specific subclass of self-similar solutions representing the so-called standard conditions and proceed to show that the NSE always has smooth solutions for all t under rescaling for both space-periodic and Schwartz-class initial data under standard parabolic conditions. Definitions, foundational concepts and examples are given in the Appendix; also in the Appendix, we apply our method to the incompressible Euler equations, demonstrating that embedding either space-periodic or Schwartz-class initial data into a carefully selected self-similar solution guarantees the flow will remain smooth for all time.

2. Applying the invariant theory of Bouton-Lie to the NSE

Bouton's work on differential invariants is considered foundational in invariant theory [7]. We summarize the theory (For an introductory text to the theory, see Ref. [8]) for the general linear differential equation (GLDE) and adapt it to the NSE using group-theoretic notions and theorem-style statements. This technique has already been applied to the NSE in part, and is well-known, e.g. the work of Lloyd [4].

The GLDE is given by

$$y^{(n)} + na_1 y^{(n-1)} + \frac{n(n-1)}{2} a_2 y^{(n-2)} ... + a_n y = 0,$$

where a_s and y are functions of x alone and $a_0 \equiv 1$.

Definition 2.1 (Finite transformation)**.** As an example, given is the point transformation

$$\begin{aligned} x_1 &= \chi(x), \\ y_1 &= y\,\psi(x), \end{aligned} \tag{2.1}$$

known as a *finite transformation*. Following Bouton's notation, x_1 and y_1 are the transformed variables, while χ and ψ are arbitrary analytic functions. For the NSE, we denote the transformed variables with primes instead, i.e. x', y'.

Definition 2.2 (Admitted transformation)**.** A differential equation is said to admit a point transformation, if this transformation leaves the equation unchanged, or, transforms it into an equation of the same form. The transformation (2.1) is the most general finite transformation the GLDE is known to admit.

Definition 2.3 (Infinitesimal transformation, GLDE)**.** Corresponding to the finite transformation (2.1), the *infinitesimal transformation* of the GLDE is the linear differential operator

$$X = \xi(x)\frac{\partial}{\partial x} + \eta(x,y)\frac{\partial}{\partial y},$$

where $\xi(x)$ and $\eta(x,y)$ are obtained by expanding the finite transformation to first order in a small parameter (generally, $\xi = \xi(x,y)$, however for the GLDE ξ is intentionally restricted to $\xi(x)$ because the independent variable transformations are taken to be independent of y). This operator is called the *symbol* of the transformation. Bouton extends X to include the coefficients of the GLDE and their derivatives,

$$X = X(x, y, \chi, \psi, a_s, a_i^{(j)}, y^{(\mu)}),$$

so that the invariance condition

$$XI = 0 \tag{2.2}$$

expresses the necessary and sufficient condition for a function I to be an absolute invariant or covariant of the GLDE.

Definition 2.4 (Absolute and relative invariants/covariants, GLDE)**.** An absolute invariant/covariant is a function $I = I(x, a_s, a_i^{(j)}, \ldots)$ with $I_1 = I$ under (2.1) or (2.2). A relative invariant/covariant is a function $R = R(x, a_s, a_i^{(j)}, \ldots)$ with $R_1 = fR$, where f is a factor determined by the transformation. Invariants do not involve y or its derivatives; but covariants do.

Definition 2.5 (Isobaric weight, GLDE)**.** Assign weights by

$$[y^{(\mu)}]^\nu \mapsto \mu\nu, \qquad [a_i^{(j)}]^l \mapsto (i+j)l.$$

A function is isobaric of weight W if each of its terms has total weight W.

Definition 2.6 (IRF)**.** An integral rational function (IRF) is either a polynomial or a ratio of polynomials.

Theorem 2.7 (Bouton, GLDE)**.** *Every absolute invariant/covariant of the GLDE is an integral rational function (IRF) homogeneous of degree* 0 *and isobaric of weight* 0*. Every relative invariant/covariant* R *is an IRF homogeneous of degree* λ *and isobaric of weight* W*, denoted* $R^{(\lambda,W)}$*.*

Corollary 2.8. *The transformation of a function is determined by the transformable entities within its functional form. This is seen in the reasoning of Bouton by the way he calculates isobaric weights and powers of homogeneity. Under finite transformation, the function* $I(x, y, y', y'', ..., y^{(\mu)}, a_1, a_2, ..., a_i^{(j)}, ...)$ *will transform according to how its individual arguments would transform within the functional form* I*. The same is true for the relative invariants/covariants* R*.*

Corollary 2.9. *For the GLDE, the factor for relative invariants is*

$$f = \frac{\psi^\lambda}{(\chi')^W}.$$

NSE scaling groups and generators

For the GLDE, Bouton used the index “1” to denote a transformed variable; for the NSE, we will use “prime” for the same purpose. We use the most general constant-parameter scaling admitted by the NSE (with gravity set to zero) [9]:

Definition 2.10 (Finite transformation, NSE scaling, general case)**.**

$$\begin{aligned} &(x', y', z') = k^{\alpha_x}(x, y, z), \qquad t' = k^{\alpha_t} t, \\ &(u', v', w') = k^{\alpha_x-\alpha_t}(u, v, w), \quad p' = k^{2(\alpha_x-\alpha_t)}p, \qquad 0 < k < \infty,\ \alpha_x, \alpha_t \in \mathbb{R}. \\ &\nu' = k^{2\alpha_x-\alpha_t}\nu, \end{aligned} \tag{2.3}$$

This is the most general finite scaling transformation admitted by the NSE when the gravity $g_i = 0$. In it, all three spatial variables transform with the same scaling exponent α_x and all three velocity components transform with the same scaling exponent $\alpha_x - \alpha_t$; α_x and α_t are arbitrary real numbers. Also, we set “p” to stand for p/ρ and thus the finite transformation for p in (2.3) becomes $p' = k^{2\alpha_x-2\alpha_t}p$.

Definition 2.11 (Finite transformation, NSE scaling, standard case)**.** The classical special case is obtained by $\alpha_x = 1,\ \alpha_t = 2$ with gravitational acceleration $g_i = 0$ (or $\alpha_t = 2\alpha_x$). We will call this case “the standard case”. An entire section of this paper is devoted to explaining the standard case, see below. The regularity results of this article are derived for the standard case.

Definition 2.12 (Infinitesimal transformation, NSE scaling, general case)**.** The infinitesimal generator corresponding to (2.3) is

$$X = \alpha_x (x\partial_x + y\partial_y + z\partial_z) + \alpha_t\, t\partial_t + (\alpha_x - \alpha_t)(u\partial_u + v\partial_v + w\partial_w) + 2(\alpha_x - \alpha_t)\, p\partial_p + (2\alpha_x - \alpha_t)\, \nu\partial_\nu.$$

Definition 2.13 (Other infinitesimal transformations of the NSE, case of $\alpha_t = 2\alpha_x$)**.** In the case of the Navier–Stokes equations, the admitted infinitesimal transformations are generated by differential operators acting on the variables (t, x, y, z, u, v, w, p). Examples include [4] the scaling operator

$$\mathcal{X} \;=\; x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z} + 2t\frac{\partial}{\partial t} - u\frac{\partial}{\partial u} - v\frac{\partial}{\partial v} - w\frac{\partial}{\partial w} - 2p\frac{\partial}{\partial p}, \tag{2.4}$$

and the infinitesimal rotations

$$\begin{aligned}
\mathcal{R}_x &= y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} + v\frac{\partial}{\partial w} - w\frac{\partial}{\partial v},\\
\mathcal{R}_y &= z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} + w\frac{\partial}{\partial u} - u\frac{\partial}{\partial w},\\
\mathcal{R}_z &= x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} + u\frac{\partial}{\partial v} - v\frac{\partial}{\partial u}.
\end{aligned} \tag{2.5}$$

Together with the time translation $\mathcal{T} = \partial/\partial t$, these operators form the infinitesimal generators of the admitted group of the NSE in the standard case.

Remark 2.14. As in Bouton's treatment of the GLDE, the condition for a function $I(u, v, w, p)$ to be an absolute invariant or covariant of the NSE is that it is annihilated by each generator. Explicitly, the system is

$$\begin{aligned}
\mathcal{T}I &= 0,\\
\mathcal{X}I &= 0,\\
\mathcal{R}_xI &= 0,\\
\mathcal{R}_yI &= 0,\\
\mathcal{R}_zI &= 0.
\end{aligned} \tag{2.6}$$

Bouton-Lie adaptation to the NSE. It is through this adaptation that we are able to benefit more fully from the theory of Lie, and move past the well-known NSE invariants [4] in the standard case.

Definition 2.15 (Absolute and relative invariants/covariants, NSE)**.** An absolute invariant/covariant is a function $I = I(x, y, z, t)$ with $I' = I$ under (2.3). A relative invariant/covariant is a function

$$R = R(x, y, z, t, u, v, w, p, \nu, \ldots)$$

such that $R' = fR$ under (2.3). Invariants do not involve (u, v, w, p); but covariants do.

Definition 2.16 (Isobaric weight, NSE)**.** Assign weights

$$W(x^a) = a\alpha_x, \quad W(t^b) = b\alpha_t, \quad W(u^a) = W(v^a) = W(w^a) = (\alpha_x - \alpha_t)a,$$
$$W(p^b) = 2(\alpha_x - \alpha_t)b, \quad W(\nu^c) = (2\alpha_x - \alpha_t)c.$$

A function is isobaric of weight W if every term has total weight W. When variables are multiplied, their weights are added.

Definition 2.17 (Integral rational function (IRF) and homogeneity)**.** An *integral rational function* (IRF) in the dependent variables (u, v, w, p) is a function that can be written as the ratio of two polynomials with integer exponents in those variables. If P and Q are the numerator and denominator polynomials, we call

$$R = \frac{P(u, v, w, p, \ldots)}{Q(u, v, w, p, \ldots)}$$

an IRF. The IRF R is said to be *homogeneous of degree* λ (in the dependent variables) if, under the uniform rescaling $(u, v, w, p) \mapsto s\,(u, v, w, p)$, one has

$$R(su, sv, sw, sp, \ldots) = s^{\lambda}R(u, v, w, p, \ldots)$$

for all $s > 0$. For an example monomial $M = u^a v^b w^c p^d$ this degree is $\lambda(M) = a + b + c + d$. For a general IRF the degree λ is defined as the degree of the numerator minus the degree of the denominator.

Definition 2.18 (Isobaric weight)**.** Under the NSE scaling (2.3) with exponents α_x, α_t, assign the weights

$$W(u) = W(v) = W(w) = \alpha_x - \alpha_t, \qquad W(p) = 2(\alpha_x - \alpha_t).$$

For an example monomial $M = u^a v^b w^c p^d$ the isobaric weight is

$$W(M) = (\alpha_x - \alpha_t)(a + b + c) + 2(\alpha_x - \alpha_t)d.$$

For a general IRF, W is the weight of the numerator minus the weight of the denominator computed termwise in this manner.

Theorem 2.19 (Bouton, NSE adaptation)**.** *Every absolute invariant/covariant of the NSE is an IRF homogeneous of degree* 0 *(in the dependent variables) and isobaric of weight* 0*. Every relative invariant/covariant R is an IRF homogeneous of degree λ and isobaric of weight W. In particular, the dependent variables carry the weights*

$$W(u) = W(v) = W(w) = \alpha_x - \alpha_t, \qquad W(p) = 2(\alpha_x - \alpha_t).$$

Moreover, for an example monomial $M = u^a v^b w^c p^d$ the following relation holds between λ and W:

$$\lambda(M) = a + b + c + d, \qquad W(M) = (\alpha_x - \alpha_t)\big(\lambda(M) + d\big).$$

Corollary 2.20. *The transformation of a function is determined by the transformable entities within its functional form. This is seen in the reasoning of Bouton by the way he calculates isobaric weights and powers of homogeneity. Under finite transformation, the function I(x,y,z,t) will transform according to how its individual arguments would transform within the functional form I. The same is true for the relative invariant functions $u(x,y,z,t)$, $v(x,y,z,t)$, $w(x,y,z,t)$, $p(x,y,z,t)$.*

Lemma 2.21 (Isobaric PDE constraints)**.** *Let $\vec{r} = (x,y,z)$. The isobarity conditions implied by* (2.3) *force the relative covariants u,v,w,p, and thus the relative invariant functions $u(x,y,z,t)$, $v(x,y,z,t)$, $w(x,y,z,t)$, $p(x,y,z,t)$ to satisfy*

$$(\vec{r}\cdot\nabla)\vec{u} + t\,\frac{\alpha_t}{\alpha_x}\,\partial_t\vec{u} = \frac{\alpha_x - \alpha_t}{\alpha_x}\,\vec{u}, \qquad (\vec{r}\cdot\nabla)p + t\,\frac{\alpha_t}{\alpha_x}\,\partial_t p = \frac{2(\alpha_x - \alpha_t)}{\alpha_x}\,p. \tag{2.7}$$

Corollary 2.22. *Under the general scaling transformation (2.3), the factor of the relative invariants/covariants of the NSE is $f = k^{(\alpha_x - \alpha_t)}$ for the velocity and $f = k^{2(\alpha_x-\alpha_t)}$ for the pressure.*

Corollary 2.23 (Form of invariants and examples)**.** *All admissible invariants and covariants in this setting are IRFs. When the denominator is constant, they reduce to isobaric polynomials. Examples include*

$$R^{(2,-2)} = u^2 + v^2 + w^2, \qquad R^{(1,2(\alpha_x-\alpha_t))} = p.$$

Comparison: GLDE vs. NSE.

Item	GLDE	NSE
Independent variables	x	(x,y,z,t)
Dependent variables	y	(u,v,w,p)
Finite transformations	$x_1 = \chi(x)$, $y_1 = y\psi(x)$	Scaling (2.3)
Infinitesimal generator	general $X = \xi\partial_x + \phi\partial_y$	generator given above
Absolute invariants	IRF, degree 0, weight 0	IRF, degree 0, weight 0
Relative invariants	$R^{(\lambda,W)}$, $f = \psi^\lambda/(\chi')^W$	$R^{(\lambda,W)}$, $f = k^W$ (per weight)
Weight assignment	$[y^{(\mu)}]^\nu \mapsto \mu\nu$, $[a_i^{(j)}]^l \mapsto (i+j)l$	$W(u) = \alpha_x - \alpha_t$, $W(p) = 2(\alpha_x - \alpha_t)$, etc.
Extra consequences	algebraic restrictions	PDE constraints (2.7)
Sample Isobaric Polynomials	$y^{(12)} + (y''')^4 + (y'')^6$	$u^2+v^2+w^2$, p

3. Self-similar solutions of the NSE

Definition 3.1 (Self-similar solutions of the NSE)**.** The relative covariants u,v,w,p, together with their corresponding relative invariant functions

$$u(x,y,z,t), \quad v(x,y,z,t), \quad w(x,y,z,t), \quad p(x,y,z,t),$$

emerge in the analysis of the scaling symmetry group of the Navier–Stokes equations. A function of this type is a solution of the NSE which also satisfies the isobaric PDE constraints of Bouton, Eq. (2.7). Such solutions are called *self-similar solutions*, since they preserve their structure under scaling transformations and therefore look the same across all scales.

The PDE constraints of Bouton Eq. (2.7) are integrated and yield:

$$\begin{aligned} \vec{u} &= t^{\frac{\alpha_x-\alpha_t}{\alpha_t}}\mathbf{F}\left(\frac{x}{t^{\frac{\alpha_x}{\alpha_t}}},\frac{y}{t^{\frac{\alpha_x}{\alpha_t}}},\frac{z}{t^{\frac{\alpha_x}{\alpha_t}}}\right)\\ p &= t^{\frac{2(\alpha_x-\alpha_t)}{\alpha_t}}F\left(\frac{x}{t^{\frac{\alpha_x}{\alpha_t}}},\frac{y}{t^{\frac{\alpha_x}{\alpha_t}}},\frac{z}{t^{\frac{\alpha_x}{\alpha_t}}}\right). \end{aligned} \tag{3.1}$$

The form (3.1) is a necessary condition that all self-similar solutions must meet. They are integral rational functions (IRF) due to the scaling invariance of the system. This means they are either isobaric polynomials or the ratio of such polynomials; such are the only functions arising in the study of the scaling invariants of the NSE [2].

The above set of self-similar solutions can be expanded by the addition of the parameters L_1, L_2, L_3, T, which rescale as x, y, z, t respectively. Then the following additional solutions are also self-similar:

$$\begin{aligned} \vec{u} &= T^{\frac{\alpha_x-\alpha_t}{\alpha_t}}\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right)\\ p &= T^{\frac{2(\alpha_x-\alpha_t)}{\alpha_t}}F\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right), \end{aligned} \tag{3.2}$$

where the factor T^a carries the isobaric weight, and the rest of the solution has zero isobarity. Instead of T, the factors can contain L_i or U, where the latter rescales as u.

4. The Standard Parabolic System

4.1. **Theorems.** We will now adapt Bouton's method to the NSE in the standard case:

Theorem 4.1 (Bouton, NSE adaptation to the standard case)**.** *Every absolute invariant/covariant of the NSE is an IRF homogeneous of degree* 0 *(in the dependent variables) and isobaric of weight* 0*. Every relative invariant/covariant R is an IRF homogeneous of degree λ and isobaric of weight W. In particular, the dependent variables carry the weights*

$$W(u) = W(v) = W(w) = \alpha_x - \alpha_t = -1, \qquad W(p) = 2(\alpha_x - \alpha_t) = -2.$$

Moreover, for an example monomial $M = u^a v^b w^c p^d$ the following relation holds between λ and W:

$$\lambda(M) = a + b + c + d, \qquad W(M) = (\alpha_x - \alpha_t)\big(\lambda(M) + d\big) = -\big(\lambda(M) + d\big).$$

Lemma 4.2 (Isobaric PDE constraints, standard case)**.** *Let $\vec{r} = (x, y, z)$. The isobarity conditions implied by* (2.3) *force the relative covariants u, v, w, p, and thus the relative invariant functions $u(x, y, z, t)$, $v(x, y, z, t)$, $w(x, y, z, t)$, $p(x, y, z, t)$ to satisfy*

$$(\vec{r}\cdot\nabla)\vec{u} + 2t\,\partial_t\vec{u} = -\,\vec{u}, \qquad (\vec{r}\cdot\nabla)p + 2t\,\partial_t p = -2\,p. \tag{4.1}$$

Corollary 4.3. *Under the standard scaling transformation (2.3), with $\alpha_x = 1, \alpha_t = 2$, or $2\alpha_x = \alpha_t$, the factor of the relative invariants/covariants of the NSE is $f = 1/k$ for the velocity and $f = 1/k^2$ for the pressure.*

4.2. **Terminology.** Since the kinematic viscosity ν appears explicitly in the NSE, every physical solution inherently depends on it. In this article, ν is treated as a constant, representing a physical quantity that is neither vanishingly small nor infinite. The term *standard NSE system* is a shorthand for setting $\alpha_x = 1, \alpha_t = 2$ in (2.3), which disallows ν from rescaling and keeps the viscosity the same at all scales. Since such behavior of the viscosity is physically relevant for Newtonian fluids, it is also referred to as *natural scaling*.

It is also termed *diffusive (parabolic)* because, under the natural scaling $\alpha_x = 1, \alpha_t = 2$, the Laplacian term $\nu\Delta\vec{u}$ remains unchanged. That is, the viscosity always acts through the same diffusion mechanism, regardless of the scale; the dissipative structure of the NSE is preserved: momentum diffuses in space in a manner analogous to the diffusion of heat in the classical heat equation. The Laplacian describes how viscosity spreads momentum, and its invariance under the standard scaling shows that diffusion operates in the same way at every scale. This is typical of Newtonian fluids, where viscosity is constant at all scales and directly responsible for smoothing the velocity field. "Parabolic" is a classification term and refers to the fact that the NSE, like the heat equation, contain a built-in diffusion mechanism: disturbances spread out in space over time at a rate proportional to $\sqrt{\nu t}$. In this sense, viscosity governs the dissipation of momentum in the same way that thermal diffusivity governs the dissipation of heat.

Therefore, by "Standard Parabolic NSE system," we mean the incompressible NSE (1.1) with constant viscosity across all scales.

4.3. **The Standard System - a Set of Outside Scaling Rules.** We must emphasize, that the term *standard conditions, standard system etc.* does not define a solution of the NSE (1.1) - rather, it is an outside-imposed set of scaling rules (which are not enforced by the NSE itself)

$$\alpha_x = 1, \quad \alpha_t = 2,$$

where entities scale as

$$\vec{u}' = k^{-1}\vec{u}, \quad p' = k^{-2}p, \quad x' = kx, \quad t = k^2 t, \quad \nu' = \nu,$$

with ν being the viscosity of the Newtonian fluid. A key feature of this definition is that the Laplacian term $\nu\Delta\vec{u}$ in the NSE remains invariant: the viscous operator has the same form at all spatial scales, so momentum diffusion is encoded in the PDE in a scale-independent way.

The NSE always contains the diffusion mechanism $\nu\Delta\vec{u}$; this operator is scale-invariant under standard scaling. Consequently, every exact NSE solution reflects viscous diffusion as it evolves, because the viscous term is structurally present in the equations regardless of the particular solution.

So in short, a "standard case" designates an environment, or framework, a set of system-level scaling rules given in Definition 2.11, Theorem 4.1, Lemma 4.2 and Corollary 4.3.

4.4. **Solutions of the Standard Parabolic System.** Note that the standard parabolic NSE system can generate both self-similar solutions as well as other solutions. Consider a set of solutions of the standard parabolic system given in (3.2). In these, if $\alpha_t = 2\alpha_x$, the solutions are self-similar; if $\alpha_t \neq 2\alpha_x$, the solutions are not self-similar. However,

- In both cases, viscosity ν is constant at all scales as the solution evolves.
- The Laplacian term $\nu\Delta\vec{u}$ always governs the diffusion of momentum in the evolution of both.
- Both are solutions of the standard parabolic NSE system and reflect the viscous dissipation, coming from a constant viscosity ν throughout all scales.

Consider the self-similar scaling, $\alpha_x = 1,\ \alpha_t = 2$. From (3.2), the velocity field is

$$\vec{u}(x,y,z,t) \ = \ T^{\frac{1-2}{2}}\,\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right) = T^{-\frac{1}{2}}\,\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right).$$

Under the standard parabolic rescaling

$$x' = kx, \quad t' = k^2 t,$$

we have $T' = k^2 T$, and hence

$$\vec{u} = (k^2 T)^{-1/2}\,\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right) = k^{-1}\vec{u}.$$

Thus the solution rescales in a self-similar way.

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Now cosider the non-self-similar scaling, $\alpha_x = 5,\ \alpha_t = 3$. From (3.2), the velocity field is

$$\vec{u}(x,y,z,t) \ = \ T^{\frac{5-3}{3}}\,\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right) = T^{\frac{2}{3}}\,\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right).$$

Under the standard parabolic rescaling

$$x' = kx, \quad t' = k^2 t,$$

we have $T' = k^2 T$, and hence

$$\vec{u} = (k^2 T)^{\frac{2}{3}} \, \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right) = k^{\frac{4}{3}} \vec{u}.$$

Here the rescaling law differs from $\vec{u}' = k^{-1}\vec{u}$, so the solution is not self-similar, though it still solves the standard parabolic NSE system with constant ν at all scales, and rescales in a standard environment.

We need to emphasize that these, second type solutions are of special importance in addressing NSE regularity and are examined in detail below. The solution is a self-similar solution of Bouton (3.2) under the mismatched condition $\alpha_x \neq 2\alpha_t$; it is a product of the standard parabolic NSE system (1.1) and rescales as a non-self-similar solution in a standard environment.

4.5. **Self-similar solutions of Leray.** The criteria of the standard NSE system $2\alpha_x = \alpha_t$, or $\alpha_x = 1, \alpha_t = 2$ isolate a particular class from the full set of the NSE self-similar solutions (3.1)- those of Leray [10]

$$\vec{u}(x, y, z, t) = \frac{1}{\sqrt{t}} \mathbf{F}\left(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}}\right)$$

$$p(x, y, z, t) = \frac{1}{t} F\left(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}}\right). \tag{4.2}$$

Such solutions are weak solutions [11], which is due to their discontinuity at $t = 0$. Previously, we showed that a subset of Leray's solutions are infinitely differentiable for $t > 0$; as well as showed the existence of a conserved, coercive quantity in standard conditions, namely $\mathcal{E}$-the cavitation number ([2], Theorem IV.2). However, this still does not allow us to assign a finite, well-behaved initial datum at $t = 0$, and the solutions cannot be considered strong.

Item	**General NSE**	**Standard NSE**
Scaling exponents	$\alpha_x, \alpha_t \in \mathbb{R}$	$\alpha_x = 1, \ \alpha_t = 2 \quad (\alpha_t = 2\alpha_x)$
Rescaling of viscosity ν	Allowed	ν is fixed at all scales
General self-sim. solutions	$\vec{u} = t^{\frac{\alpha_x - \alpha_t}{\alpha_t}} \mathbf{F}\left(\frac{x}{t^{\alpha_x/\alpha_t}}, \frac{y}{t^{\alpha_x/\alpha_t}}, \frac{z}{t^{\alpha_x/\alpha_t}}\right)$, $p = t^{\frac{2(\alpha_x - \alpha_t)}{\alpha_t}} F\left(\frac{x}{t^{\alpha_x/\alpha_t}}, \frac{y}{t^{\alpha_x/\alpha_t}}, \frac{z}{t^{\alpha_x/\alpha_t}}\right)$	$\vec{u} = \frac{1}{\sqrt{t}} \mathbf{F}\left(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}}\right)$, $p = \frac{1}{t} F\left(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}}\right)$
General case	No rescaling parameters	Same
Extended case	Rescaling par. L, L_1, L_2, L_3, T, U, etc.	Same
Extended self-sim. solutions	$\vec{u} = T^{\frac{\alpha_x - \alpha_t}{\alpha_t}} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$, $p = T^{\frac{2(\alpha_x - \alpha_t)}{\alpha_t}} F\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$	$\vec{u} = \frac{1}{\sqrt{T}} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$, or $U \, \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$, or $\frac{1}{L} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$, or $\frac{L}{T} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$

By definition, a strong solution to the NSE is one that is not only smooth (or regular) for positive times but also has initial data continuous up to $t = 0$. Even if a subset of Leray's self-similar solutions in 3-D (4.2) are infinitely differentiable for all $t > 0$, the singular behavior at $t = 0$ prevents them from being well-defined at the initial time. In other words, the singularity at $t = 0$ means that the initial data is too irregular—hence, despite their smoothness for $t > 0$, they are classified as weak solutions because they only satisfy the NSE in a distributional sense at $t = 0$ and fail to meet the criteria for strong solutions on the entire time interval including $t = 0$.

To avoid the discontinuity at $t = 0$ of Leray's self-similar solutions, a common practice is to use a subset of the self-similar solutions (3.2). These are Bouton's self-similar solutions for standard conditions $\alpha_t = 2\alpha_x$ or $\alpha_x = 1, \alpha_t = 2$:

$$
\begin{aligned}
\vec{u} &= \frac{1}{\sqrt{T}}\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \\
\vec{u} &= U\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \\
\vec{u} &= \frac{1}{L}\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \text{ or} \\
\vec{u} &= \frac{L}{T}\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right),
\end{aligned}
\tag{4.3}
$$

where the required isobaric weight of $\vec{u}$, equal to ($\alpha_x - \alpha_t = -1$) is achieved through the rescaleable parameters L, L_1, L_2, L_3, T, U, which scale as $(x, y, z), x, y, z, t, u$ accordingly, while $\mathbf{F}$ is an IRF of weight zero. $\mathbf{F}$ is composed of either isobaric polynomials, or the ratio of such polynomials. We have listed the solutions (4.3) for the sake of completeness, however they are still subject to supercriticality (see below), which makes them unfit in addressing the regularity of the NSE.

4.6. **Supercriticality.** The standard NSE system is also characterized by *energy-supercriticality* [2], [12]—a condition that is often cited as an obstruction to proving global regularity of solutions. While (4.3) are smooth at $t = 0$, they are self-similar under the standard rescaling $\alpha_x = 1, \alpha_t = 2$, which corresponds to an energy-supercritical regime [2]. Our goal is to avoid supercriticality and have solutions, which rescale in a desired manner, as described in the next section.

5. Mismatched Self-Similar Solutions Under Standard Conditions

The most general finite scaling transformation admitted by the NSE is found in (2.3). It depends on the real-valued scaling exponents α_x, α_t. That is, the scaling of the dependent varialbles u, v, w, p is determined by the scaling of the space and time variables x, y, z, t. Setting $\alpha_x = 1, \alpha_t = 2$ results in the *standard scaling*, as defined in Section 4 above. It mandates how the spatial variables x, y, z and the temporal variable t are to rescale, and thus define the rescaling of the fluid environment. Being a parameter of the NSE, the viscosity ν remains the same at all scales in this specific case.

Within the extended set of Bouton's self-similar solutions (3.2), some rescale in a self-similar fashion in such standard scaling environment: these are all self-similar solutions which rescale with $\alpha_t = 2\alpha_x$. We will denote all remaining self-similar solutions with

$$
\begin{aligned}
\vec{u} &= T^{\frac{\beta_x - \beta_t}{\beta_t}}\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right) \\
p &= T^{\frac{2(\beta_x - \beta_t)}{\beta_t}}F\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \text{or} \\
\vec{u} &= U\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \\
\vec{u} &= \frac{L}{T}\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right),
\end{aligned}
\tag{5.1}
$$

where U, L, T and other entities are chosen to rescale according to $\beta_t \neq 2\beta_x$. These solutions are of crucial importance in addressing the regularity of the NSE. At $t = 0$, (5.1) have the form

$$\begin{aligned} \vec{u}_0 &= T^{\frac{\beta_x-\beta_t}{\beta_t}} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, 0\right) \\ p_0 &= T^{\frac{2(\beta_x-\beta_t)}{\beta_t}} F\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, 0\right). \end{aligned} \tag{5.2}$$

Notice that in (5.1) some solutions have a strictly defined scaling behavior, e.g. these where the prefactor T is raised to the exponent $(\beta_x - \beta_t)/\beta_t$. While others do not contain a scaling exponent and can thus behave as self-smilar solutions under both standard and non-standard rescalings. In practice, many well-known solutions to the NSE, such as those describing Couette flow [13]

$$u = u_0\left(\frac{y}{h} - \frac{2}{\pi}\sum \frac{1}{n} e^{-n^2\pi^2\frac{\nu t}{h^2}} \sin\left[n\pi\left(1 - \frac{y}{h}\right)\right]\right)$$

or the Taylor-Green vortex [14],

$$\begin{aligned} u_i &= u_{0i} F_{0i}(x, y, z), \text{ initial motion}, i = 1, 2, 3 \\ u_i &= \sum u_{0i} F_{0i}(x, y, z, t), \text{ approximations 1-3}, i = 1, 2, 3 \end{aligned}$$

achieve the isobarity required by Bouton for all self-similar solutions (3.1) through velocity parameters (e.g. u_{0i}), which scale the same way as u. The rest of the function is an isobaric polynomial with zero weight F_{0i} (zero isobarity functions do not rescale), which ensure that the solutions vanish at infinity or remain periodic.

Lemma 5.1. *Any function $f(x, y, z)$ can be embedded into a self-similar function of the form*

$$p_4 f\left(\tfrac{x}{p_1}, \tfrac{y}{p_2}, \tfrac{z}{p_3}\right)$$

at the identity scale.

Proof. Write any function as

$$f(x, y, z) = C_4 f\left(\tfrac{x}{C_1}, \tfrac{y}{C_2}, \tfrac{z}{C_3}\right),$$

with fixed constants $C_i = 1$. For arbitrary constants C_i, this can be rewritten as

$$p_4 f\left(\tfrac{x}{p_1}, \tfrac{y}{p_2}, \tfrac{z}{p_3}\right),$$

with $p_i = C_i$. Rescaling of the parameters yileds $p_i' = k^a p_i, k > 0$, and since we seek $p_i' = C_i$, it follows that $k = 1$, or $a = 0$, which is the trivial, or identity scaling. Thus, every non-self-similar function can be viewed as embedded in a self-similar function at the identity scale. $\square$

Corollary 5.2. *Every solution $f(x, y, z, t)$ of the NSE can be embedded in a self-similar solution of the form*

$$p_5 f\left(\tfrac{x}{p_1}, \tfrac{y}{p_2}, \tfrac{z}{p_3}, \tfrac{t}{p_4}\right)$$

at the identity scale. Hence, the self-similar solutions—isobaric polynomials or rational fields—constitute the native solutions *of the NSE, dictated directly by the symmetry and structure of the equations.*

Theorem 5.3 (Smoothness of the extended self-similar solution forms (5.1))**.** *Given are the rescaling parameters L, L_1, L_2, L_3, T and (optionally) U, which scale as x, y, z, t and (u, v, w). Let $\mathbf{F} : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$ be a vector-valued IRF of the similarity variables*

$$(\xi, \eta, \zeta, \tau) = \left(\tfrac{x}{L_1}, \tfrac{y}{L_2}, \tfrac{z}{L_3}, \tfrac{t}{T}\right),$$

whose numerator and denominator are isobaric polynomials of weight 0*. Define* $\vec{u}$ *by one of the standard Bouton forms (5.1) (and analogously, the pressure):*

$$\begin{aligned} \vec{u} &= T^{\frac{\beta_x-\beta_t}{\beta_t}}\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right) \\ p &= T^{\frac{2(\beta_x-\beta_t)}{\beta_t}}F\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right), \end{aligned}$$

where the pre-factor may involve T, U, L*, etc.*

Assume:

(E) *(*Embedding*) By Lemma* (5.1) *and Corollary* (5.2)*, at the identity scale the extended set of self-similar functions of Bouton (5.1), namely* $T^{\frac{\beta_x-\beta_t}{\beta_t}}\mathbf{F}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right)$*, ... etc. embeds any NSE solution.*

(C) *(*Clay initial regularity*) The embedded initial data are smooth on* $\mathbb{R}^3$ *and do not grow at infinity when* $t = 0$ [1]*. This is an apriori condition and it is required in order to guarantee the initial datum is physically meaningful. Because the NSE do not include a forcing term, the initial velocity is the sole energy source of the flow. Thus, in the initial moment the system is guaranteed to have finite energy*

$$\int_{\mathbb{R}^3} |\vec{u}(x,y,z,0)|^2\,dx\,dy\,dz < \infty.$$

Then there exist velocity and pressure fields $\vec{u}, p$ *from the set* (5.1) *such that* $\vec{u}, p$ *are smooth IRFs on* $\mathbb{R}^3 \times \mathbb{R}$ *and* $\mathbb{R}$ *respectively, with denominators that are* nowhere zero*. Consequently,* $\vec{u}, p$ *are globally smooth and do not grow at infinity. These will be the solutions used to address the regularity of the NSE.*

Proof. By the Clay initial regularity assumption (C), the embedded initial data are smooth on $\mathbb{R}^3$ at $t = 0$. Thus $\mathbf{F}(\xi,\eta,\zeta,0)$ is smooth for all $(\xi,\eta,\zeta) \in \mathbb{R}^3$. Consequently $Q(\xi,\eta,\zeta,0)$ cannot vanish anywhere unless the zero is removable by cancellation with $\mathbf{P}$. Cancelling all common factors (which preserves the IRF nature and weight 0), we may and do assume that $Q(\xi,\eta,\zeta,0)$ is *nowhere zero.*

Now suppose $\mathbf{F}_i = \mathbf{P}_i/Q_i,\ i = 1,2,3$ where $\mathbf{P}$ is a vector of isobaric polynomials of weight 0 and Q_i is an isobaric polynomial of weight 0. Any zero $(\xi_0,\eta_0,\zeta_0,0)$ of Q_i would correspond to some point $(x_0,y_0,z_0,0)$ in physical variables at which $\vec{u}$ (hence $\mathbf{F}$) blows up, unless the zero is *removable* (i.e. a common factor cancels). That is, denominator zeros would force an algebraic singularity somewhere in $(x,0)$. This would certainly affect the total initial energy of the system, in contradiction with condition (C). Thus, the assumption that the denominator vanishes anywhere at $t = 0$, is inadmissible.

To ensure that the solutions do not grow at infinity when $t > 0$, we need multiplier functions such as $\exp(-(x^2+y^2+z^2)/L^2)$, $\exp(-t/T)$ etc. The extended set of Bouton (5.1) is able to supply such solutions, because the multiplier functions are of isobaric weight zero. Alternatively, they may be space-periodic trigonometric functions, as described above in the cases of Couette flow and Taylor-Green vortex. Additionally, any Taylor expansion of a smooth, closed-form function in the similarity variables is zero-weight isobaric polynomial, and as such it will exist in the set of Bouton (5.1).

Therefore, after removing any common polynomial factors between $\mathbf{P}_i$ and Q_i (which does not change $\mathbf{F}$), we have shown that we can always choose our solution from the set (5.1) so that Q_i has no zeros on $\mathbb{R}^3 \times \mathbb{R}$. Each of our chosen $\vec{u}, p$ are IRFs with smooth functional form (no poles, zeros, or discontinuities). Additionally, they have been so chosen, that they do not grow at infinity. Hence $\mathbf{F}$ can always be chosen out of the set (5.1) as smooth on all similarity variables. These are the solutions we will use in addressing the regularity of the NSE. □

6. The Beale-Kato-Majda (BKM) Criterion

This section introduces the BKM Criterion, originally published in Ref.[15], while a clearly written derivation of BKM can be found also here [16].

Theorem 6.1 (BKM Criterion)**.** *The BKM criterion states that a smooth solution of the NSE (or Euler) equations can only lose regularity (blow up) if the vorticity grows without bound. In particular, let $u(x,t)$ be a smooth solution of the incompressible NSE equations on a time interval $[0,T)$. Define the vorticity*

$$\omega(x,t) = \nabla \times u(x,t).$$

If

$$\int_0^T \sup_{x\in\mathbb{R}^3} |\omega(x,t)|\, dt < \infty,$$

then the solution $u(x,t)$ can be extended smoothly beyond time T. Equivalently, the only possible mechanism for a finite–time blowup is

$$\sup_{x\in\mathbb{R}^3} |\omega(x,t)| \;\to\; \infty \quad \textit{as } t \to T.$$

The self-similar solutions of Bouton (5.1), being globally smooth by Theorem 5.3, play a central role in addressing the regularity of the NSE. Their advantage lies in the explicit functional form $\mathbf{F}$: smooth isobaric rational functions (IRFs), either polynomials or smooth ratios of polynomials. Thus blowup cannot occur through discontinuities in $\mathbf{F}$.

The only remaining concern is uncontrolled rescaling within the classical NSE, which manifests as a blowup mechanism (or, *scaling-induced blowup*). Once fine-scale control is imposed, this instability is removed and the blowup mechanism is eliminated.

Application of the BKM criterion. Since the vorticity consists of spatial derivatives of the velocity field,

$$\omega = \nabla \times u \;\subset\; \{\tfrac{\partial u}{\partial x_i}\},$$

we analyze the behavior of these derivatives under the Bouton rescaling. As long as they remain bounded for all t, the BKM criterion ensures that finite-time blowup cannot occur.

Lemma 6.2. *Consider the smooth (as per Theorem 5.3) self-similar solution (5.1)*

$$\vec{u}^* = T^{\frac{\beta_x-\beta_t}{\beta_t}} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right),$$

with isobaric weight $\beta_x - \beta_t$. Under standard scaling ($\alpha_x = 1,\ \alpha_t = 2$), the solution is free from blowup in velocity, energy, and vorticity provided that

$$\frac{\beta_x}{\beta_t} > \tfrac{3}{2}.$$

Proof. The native scaling of the solution is expressed via the scaling exponents β_x, β_t; the background standard scaling (the scaling of the environment) is expressed through α_x, α_t, where $\alpha_x = 1, \alpha_t = 2$. A Bouton solution of the form (5.1) is smooth as per Theorem 5.3. The scaling of vorticity depends on the derivatives:

$$\left(\frac{\partial \vec{u}^*}{\partial x}\right)' = \left(\frac{\partial (\vec{u}^*)'}{\partial x'}\right) = \frac{1}{k^{\alpha_x}}\frac{\partial}{\partial x}\left(T'^{\frac{\beta_x-\beta_t}{\beta_t}}\mathbf{F}'\right) = \frac{1}{k^{\alpha_x}}\frac{\partial}{\partial x}\left((k^{\alpha_t}T)^{\frac{\beta_x-\beta_t}{\beta_t}}\mathbf{F}'\right) = \frac{1}{k^{\alpha_x}}\frac{\partial}{\partial x}\left((k^{\alpha_t}T)^{\frac{\beta_x-\beta_t}{\beta_t}}\mathbf{F}\right),$$

notice that since $\mathbf{F}$ is of zero isobarity, $\mathbf{F}' = \mathbf{F}$, and $\mathbf{F}$ does not rescale. Because $\alpha_x = 1, \alpha_t = 2$, we get

$$\left(\frac{\partial \vec{u}^*}{\partial x}\right)' = k^{\frac{2\beta_x - 3\beta_t}{\beta_t}}\left(\frac{\partial \vec{u}^*}{\partial x}\right).$$

The scaling of the energy in standard conditions is

$$E' = \int_{\mathbb{R}^3} |\vec{u}^{*\prime}|^2 dx'dy'dz' = \int_{\mathbb{R}^3} |T'^{\frac{(\beta_x-\beta_t)}{\beta_t}}\mathbf{F}'|^2 dx'dy'dz' = \int_{\mathbb{R}^3} |(k^{\alpha_t}T)^{\frac{(\beta_x-\beta_t)}{\beta_t}}\mathbf{F}|^2 (k^{3\alpha_x}) dxdydz;$$

bringing the factors outside of the integral and recalling that $\alpha_x = 1, \alpha_t = 2$, we get

$$E' = k^{\frac{4\beta_x - \beta_t}{\beta_t}} E.$$

The scaling of the velocity itself in standard conditions is

$$\vec{u}^{*\prime} = T'^{\frac{(\beta_x-\beta_t)}{\beta_t}}\mathbf{F}' = (k^{\alpha_t}T)^{\frac{(\beta_x-\beta_t)}{\beta_t}}\mathbf{F} = k^{\alpha_t\frac{(\beta_x-\beta_t)}{\beta_t}}T^{\frac{(\beta_x-\beta_t)}{\beta_t}}\mathbf{F} = k^{\alpha_t\frac{(\beta_x-\beta_t)}{\beta_t}}\vec{u}^{*},$$

$$\vec{u}^{*\prime} = k^{\frac{2(\beta_x-\beta_t)}{\beta_t}}\vec{u}^{*}.$$

All three quantities remain bounded under scaling if $\beta_x/\beta_t > 3/2$. The case becomes energy-subcritical. For numerical illustrations, see the Appendix F. □

7. Existence, and Smoothness Theorems

In this section, we adapt the well-known Theorems of existence and smoothness of Fujita-Kato and Leray Hopf to the study and analysis of Bouton's self-similar solutions.

Theorem 7.1 (Fujita–Kato [1], [17] - [21])**.** *Let $u_0 : \mathbb{R}^3 \to \mathbb{R}^3$ be a divergence-free vector field, i.e.*

$$\nabla \cdot u_0(x) = 0 \quad \text{for all } x \in \mathbb{R}^3.$$

Assume that u_0 satisfies one of the following conditions:

(A) *u_0 is integrable in the cubic sense,*

$$\int_{\mathbb{R}^3} |u_0(x)|^3\,dx < \infty,$$

(B) *u_0 and its derivatives up to some order $s > \frac{3}{2}$ are square-integrable,*

$$\int_{\mathbb{R}^3} \left|D^\alpha u_0(x)\right|^2 dx < \infty \quad \text{for all multi-indices } \alpha \text{ with } |\alpha| \le s,$$

Note: For a multi-index $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ with nonnegative integers α_i, we write

$$D^\alpha u(x) = \frac{\partial^{|\alpha|}u(x)}{\partial x_1^{\alpha_1}\,\partial x_2^{\alpha_2}\,\partial x_3^{\alpha_3}}, \qquad |\alpha| = \alpha_1 + \alpha_2 + \alpha_3.$$

For example, if $\alpha = (1,0,0)$ then $D^\alpha u = \frac{\partial u}{\partial x_1}$, while if $\alpha = (0,2,1)$ then $D^\alpha u = \frac{\partial^3 u}{\partial x_2^2 \partial x_3}$.

(C) *u_0 is smooth and rapidly decays at infinity together with all of its derivatives, i.e.*

$$\sup_{x\in\mathbb{R}^3} (1+|x|)^N \left|D^\alpha u_0(x)\right| < \infty \quad \text{for all integers } N \ge 0 \text{ and all multi-indices } \alpha,$$

(D) *u_0 is smooth and periodic*

Consider the incompressible NSE (1.1)

$$\begin{cases} \partial_t u + (u\cdot\nabla)u = \nu\Delta u - \nabla p, & \\ \nabla\cdot u = 0, & x \in \mathbb{R}^3\ ,\ t > 0, \\ u(x,0) = u_0(x), & \end{cases}$$

with viscosity $\nu > 0$. Then:

(1) *(*Local existence, uniqueness, and energy boundedness*) There exists $T > 0$ such that there is a unique solution $u(x,t)$ defined for $0 \le t < T$, which becomes smooth for all $t > 0$ up to the maximal existence time T. Moreover, if the initial data u_0 has finite energy,*

$$\int_{\mathbb{R}^3} |u_0(x)|^2\,dx < \infty,$$

then the solution satisfies

$$\int_{\mathbb{R}^3} |u(x,t)|^2\,dx < \infty \quad \text{for all } t \in [0,T),$$

so the kinetic energy remains bounded throughout the existence interval.

(2) *(*Persistence of structure*) The solution remains in the same class as the initial data and preserves any symmetries of u_0 (e.g. periodicity, decay, self-similarity).*
(3) *(*Global existence for small data*) If u_0 is sufficiently small in the above sense (for example, the cubic integral $\int |u_0|^3$ is small), then the solution exists smoothly for all $t > 0$.*
(4) *(*Breakdown criterion for large data*) If $T^* < \infty$ is the maximal time of existence, then*

$$\limsup_{t\to T^*} \int_{\mathbb{R}^3} |u(x,t)|^3 \, dx = \infty.$$

For the purposes of this present work, the above theorem can be considered to sum up to the following: *For any smooth, divergence-free initial velocity field, there exists a unique solution of the NSE that remains smooth for a finite time interval after the initial moment. More precisely, there exists a time $T > 0$ (depending on the size of the initial data) such that*

$$u(x,t) \quad \textit{is smooth for all } 0 < t < T.$$

In particular, singularities cannot appear instantly from smooth data: the flow always remains regular for at least some short time.

Theorem 7.2 (Leray–Hopf [1], [17] - [21])**.** *Let the initial velocity field $u_0 : \mathbb{R}^3 \to \mathbb{R}^3$ satisfy:*

$$\int_{\mathbb{R}^3} |u_0(x)|^2 \, dx < \infty, \qquad \nabla \cdot u_0(x) = 0.$$

Then there exists a global-in-time velocity field $u(x,t)$ such that:
(1) ***(Finite energy for all times)*** *For every $t \geq 0$,*

$$\int_{\mathbb{R}^3} |u(x,t)|^2 \, dx < \infty, \qquad \int_0^t \int_{\mathbb{R}^3} \sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{\partial u_i}{\partial x_j}(x,s) \right)^2 dx \, ds < \infty.$$

(2) ***(Weak form of the equations)*** *For every smooth test vector field $\varphi(x,t)$ with compact support and $\nabla \cdot \varphi = 0$,*

$$\int_0^\infty \int_{\mathbb{R}^3} \left(\sum_{i=1}^3 u_i(x,t) \frac{\partial \varphi_i}{\partial t}(x,t) + \sum_{i=1}^3 \sum_{j=1}^3 u_i(x,t) \frac{\partial u_i}{\partial x_j}(x,t) \, \varphi_j(x,t) \right) dx \, dt$$

$$+ \nu \int_0^\infty \int_{\mathbb{R}^3} \sum_{i=1}^3 \sum_{j=1}^3 \frac{\partial u_i}{\partial x_j}(x,t) \, \frac{\partial \varphi_i}{\partial x_j}(x,t) \, dx \, dt + \int_{\mathbb{R}^3} u_0(x) \cdot \varphi(x,0) \, dx = 0.$$

(3) ***(Energy inequality)*** *For all $t \geq 0$,*

$$\frac{1}{2} \int_{\mathbb{R}^3} |u(x,t)|^2 \, dx + \nu \int_0^t \int_{\mathbb{R}^3} \sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{\partial u_i}{\partial x_j}(x,s) \right)^2 dx \, ds \; \leq \; \frac{1}{2} \int_{\mathbb{R}^3} |u_0(x)|^2 \, dx.$$

This inequality expresses that the kinetic energy of the fluid cannot grow with time, and decreases according to the viscous dissipation term.

For more information, please see the included comparison table for the Theorems of Fujita-Kato and Leray-Hopf. Here we introduce a Lemma, which is a direct application of the Fujita–Kato theorem to smooth, periodic, or Schwartz-class data:

Corollary 7.3 (Smooth, periodic or Schwartz initial data)**.** *Let $u_0 : \mathbb{R}^3 \to \mathbb{R}^3$ be a divergence-free vector field that is either:*

- *Smooth and periodic in space, or*
- *Smooth and rapidly decaying at infinity (Schwartz class).*

Then, by the Fujita–Kato theorem, there exists a time $T > 0$ and a unique solution $u(x,t)$ of the incompressible NSE on $(0,T)$ such that:

(1) *$u(x,t)$ is smooth, space-periodic or Schwartz class accordingly for all $t \in (0,T)$ (i.e., a classical solution),*
(2) *The kinetic energy is finite at all times $t \in (0,T)$:*

$$\int_{\mathbb{R}^3} |u(x,t)|^2\, dx < \infty \quad \textit{for all } t \in (0,T).$$

For smooth initial data, the Fujita–Kato theorem guarantees strong solutions *of the same kind.*

	Leray–Hopf (1934)	**Fujita–Kato (1964)**
Initial data	Any divergence-free velocity field with finite total energy (no smoothness required).	A divergence-free velocity field that is smooth enough to have well-defined derivatives of a certain order.
Existence of solution	A solution always exists for all times, but only in a weak sense.	A solution exists at least for a short time after the initial moment and has bound energy.
Uniqueness	Not known: there may be several weak solutions for the same initial state.	The solution is unique while it exists.
Regularity	One only knows the energy of the solution is finite, not that it is smooth.	The solution is smooth for all positive times and has bound energy up to time T.
Long-time behavior	A solution exists for all time, but it might not be unique or smooth.	If the initial data is small enough, the solution exists for all time and remains smooth.
Energy law	The total kinetic energy never increases; it decreases because of viscosity.	Same property holds.

TABLE 1. Comparison of the theorems of Leray–Hopf and Fujita–Kato.

Remark 7.4. Given smooth, divergence-free initial data u_0, Lemma 5.1 shows we can always use u_0 to construct a self-similar family U_0 such, that u_0 is embedded as the $k = 1$ member of U_0. We can then use U_0 as the initial condition for the NSE Cauchy problem. By the Fujita–Kato theorem, there exists a unique, smooth solution $U(x,t;k)$ evolving from U_0. Because U_0 is self-similar in k, the full solution U is also self-similar. $U(x,t;k=1)$ recovers the physical solution with initial data u_0, and inherits all smoothness properties established for the family U.

8. Global Regularity via Self-Similarity

In this section, we present a theorem and its proof in a summary form. It represents three separate cases: space-periodic, smooth, divergence-free initial datum; Schwartz-class, smooth, divergence-free initial datum; and an arbitrary smooth, divergence-free one. Their corresponding complete theorems A.1, B.1 and corollary B.2 and their proof are given in the Appendix.

Theorem 8.1 (Global Regularity via Self-Similarity)**.** *Let $u_0(x,y,z)$ be any smooth, divergence-free, space-periodic initial datum or such of Schwartz-class. Then, under standard conditions there exists a well-posed solution (u,p), which is unique, coincides with the NSE evolution of u_0 on $[0,\infty)$, remains C^∞ for all $t \geq 0$, and has all velocity, energy, and vorticity norms uniformly bounded.*

Proof.

- **Embedding of the initial data.** By Lemma 5.1 and Corollary 5.2, we can use the given smooth, divergence-free initial field $u_0(x, y, z)$ and build a one-parameter self-similar family U_0 such, that $u_0(x, y, z)$ is realized as the $k = 1$ identity scale member of U_0. When building U_0 and its corresponding initial pressure, we make sure they belong to the set (5.2).
- **Existence and uniqueness.** Invoking the Kato–Fujita Theorem 7.1 on the initial datum U_0 of the self-similar form (5.2), one obtains a unique, smooth NSE solution (U, P) on an interval $[0, \tau)$ having the self-similar form (5.1). The self-similar solution (U, P), being a unique solution, coincides with the NSE evolution of U_0 on $[0, \tau)$. The solution (u, p), being a unique solution, coincides with the NSE evolution of u_0 on $[0, \tau)$. (u, p) is the $k = 1$ identity scale member of (U, P).
- **A priori norm estimates.** Lemma 6.2 provides exact scaling exponents for the velocity, energy and vorticity norms of (U, P) under the standard scaling $(\alpha_x = 1, \alpha_t = 2)$. It shows that for $\beta_x/\beta_t > 3/2$, none of these norms grows with the scaling parameter, yielding uniform bounds for all t. Therefore, the same applies for its $k = 1$ identity scale member (u, p).
- **Blow-up exclusion.** The Beale–Kato–Majda criterion [15] is applied to (U, P): since
$$\int_0^T \|\omega(\cdot, \tau)\| \, d\tau < \infty$$
by the uniform bound on $\|\omega\|$, no finite-time singularity can occur. This establishes smoothness of (U, P) on $[0, \infty)$ and thus smoothness of (u, p) on $[0, \infty)$.
- **Conclusion of Theorem 8.1** Together, these steps show that the $k = 1$ self-similar member is well-posed: an unique, globally smooth, space-periodic or Schwartz-class solution of the incompressible NSE evolving from u_0, with all relevant norms remaining bounded for all $t \geq 0$. We can add that large-scale blowup $k \to \infty$ is also avoided by considering the fact that the energy of the system is bounded (from Corollary 7.3 and the BKM criterion) or by placing the flow in a bounded domain.

□

9. Limiting Cases of the Model

We now examine two limiting cases of the NSE, obtained by modifying or removing the viscosity term.

9.1. The 3D Euler Equations. Setting $\nu = 0$ in the incompressible NSE yields the *Euler equations*:

$$\begin{cases} \partial_t u + (u \cdot \nabla) u = -\nabla p, \\ \nabla \cdot u = 0, \\ u(x, 0) = u_0(x). \end{cases}$$

Here $u : \mathbb{R}^3 \times [0, \infty) \to \mathbb{R}^3$ is the velocity field, and $p : \mathbb{R}^3 \times [0, \infty) \to \mathbb{R}$ is the pressure. For the Euler system, the BKM criterion [15] shows that a smooth solution u can only blow up if the vorticity $\omega = \nabla \times u$ satisfies

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty} \, dt = \infty$$

for some $T > 0$. Thus, proving smoothness depends strongly on vorticity control. Using the available BKM criterion, we have utilized Bouton's method to prove regularity of the 3-D incompressible Euler equations by ensuring the energy and vorticity norms are bound at fine scales; for full details, see Appendix G.

9.2. The 3D Burgers Equations. Unlike the NSE and Euler, Burgers equations do not enforce the incompressibility constraint. We distinguish two cases depending on the presence of viscosity.

9.2.1. *Viscous 3D Burgers Equation.* The viscous Burgers equation is given by

$$\partial_t u + (u \cdot \nabla)u = \nu \, \Delta u, \qquad u(x, 0) = u_0(x).$$

Here the diffusive term smooths out the sharp gradients that nonlinear advection alone would produce, preventing finite-time blowup and guaranteeing globally smooth solutions. As a result, for smooth initial data and $\nu > 0$, global smooth solutions are proven to exist [22]. Thus, even without incompressibility and without a BKM-type vorticity control, the $\nu\Delta$ term alone ensures global regularity of the viscous 3D Burgers equation.

Is Bouton's method applicable to the viscous Burgers equation? Since it embeds initial data into a self-similar family to protect against classical infinite rescaling and blowup, it can indeed be applied in this case. However, the BKM criterion is inapplicable here because incompressibility is not enforced. Bouton's embedding does not alter the PDE dynamics; it only constructs particular self-similar solutions. It therefore cannot show that *all* solutions (or all embedded initial data) avoid blowup unless there exists an independent PDE mechanism providing control—such as Biot–Savart plus BKM in the incompressible NSE case, or diffusion in the viscous Burgers equation. In short, applying Bouton's method in the absence of a BKM-type criterion does not, by itself, establish global regularity.

9.2.2. *Inviscid 3D Burgers Equation.* When $\nu = 0$, one obtains the inviscid Burgers equation:

$$\partial_t u + (u \cdot \nabla)u = 0.$$

This is a purely advective transport equation. Without viscosity or incompressibility, the solution typically ends up with finite-time singularities, such as gradient blowup or shock formation, even for smooth initial data: the velocity field ceases to remain smooth beyond the shock time. This phenomenon, well established in 1D [23], extends to higher dimensions.

Bouton's framework embeds solutions into self-similar families, thereby preventing uncontrolled rescaling that could otherwise lead to blowup. In the inviscid Burgers equation, this embedding can still be constructed for smooth, periodic or Schwartz initial data, producing self-similar solution forms. Although again, unlike the NSE system, no auxiliary control principle such as Biot–Savart plus BKM is available: the Burgers dynamics lack both incompressibility and viscous dissipation. Consequently, the embedded forms remain subject to finite-time gradient blowup (shock formation), a phenomenon rigorously established for 1D and extended to higher dimensions. Thus, while Bouton's method organizes the solution space into rescaling-invariant families, it does not in this case prevent the occurrence of singularities or yield a regularity result.

9.3. **Takeaways.**

- For the 3D Euler system ($\nu = 0$, divergence-free), regularity is conditional on the BKM vorticity criterion. Bouton's self-similar solutions provide candidate globally smooth embeddings consistent with this conditional picture.
- For the viscous 3D Burgers system, the diffusion term $\nu\Delta u$ regularizes solutions and enforces global smoothness, with monotone decay of kinetic energy. Since incompressibility and vorticity dynamics are absent, a BKM-type criterion is unavailable, and Bouton's method cannot yield a regularity statement here.
- For the inviscid 3D Burgers system, the absence of both incompressibility and viscosity leads to finite-time shock formation, even from smooth initial data. Global smooth regularity is therefore impossible. Without a BKM framework to control vorticity, Bouton's method is again inapplicable.

The limiting cases of the model illustrate the precise role of each structural ingredient in establishing or failing to establish regularity. For the Euler system ($\nu = 0$, divergence-free), incompressibility channels the dynamics through vorticity, and the BKM criterion provides conditional control: blowup can only occur if the vorticity norm diverges. Once the exact functional form of solutions is identified as IRFs, the sole remaining blowup mechanism is uncontrolled rescaling, which Bouton's embedding eliminates, thereby closing the Euler case under the BKM framework. In contrast, for the viscous Burgers system, even without incompressibility or vorticity control, the strong diffusion term $\nu\Delta u$ alone ensures global smoothness. This shows that viscosity can suffice in simpler flows, though it is not strong enough by itself to settle the NSE problem. At the opposite extreme, the inviscid Burgers system lacks both incompressibility and viscosity; its purely advective dynamics produce finite-time shocks from smooth data, making global smooth regularity impossible and underscoring the necessity of additional structural controls. By comparison, the full NSE combine all favorable ingredients: incompressibility, BKM vorticity control, and viscosity. Bouton's method then eliminates the final instability due to rescaling, ensuring bounded vorticity and thus completing the case for global regularity. In this way, the limiting cases do not merely provide contrasts, but actively foster the regularity proof for NSE by isolating and clarifying the contribution of each mechanism. This paragraph is summed up in the following

Proposition 9.1. *The comparison of the limiting cases highlights the distinct contributions of each structural mechanism.*

- *In the Euler equations, incompressibility and the BKM vorticity criterion provide conditional regularity, and Bouton's method eliminates the rescaling instability.*
- *In the viscous Burgers equations, viscosity alone regularizes solutions, though without incompressibility or BKM control.*
- *In the inviscid Burgers equations, the absence of both viscosity and incompressibility leads to shock formation in finite time.*

Therefore, the full NSE, which combine viscosity, incompressibility, and the BKM vorticity criterion, are fully stabilized once Bouton's method rules out infinite rescaling. In this setting, vorticity remains bounded, and global smooth regularity follows.

10. Comparison with Classical and Modern Results

The global regularity result presented above can be viewed in the context of several landmark contributions to the study of the NSE:

- **Leray's Weak Solutions and Small-Data Smoothness.** In his foundational work [10], Leray constructed global–in–time weak solutions for arbitrary finite-energy initial data, but could only prove partial regularity in three dimensions. Subsequent refinements (e.g. Kato–Fujita [24], Temam [18]) established that sufficiently small initial data yield global smooth (C^∞) solutions. The present self-similar framework extends beyond these small-data regimes to encompass arbitrarily large, smooth, divergence-free initial data, while still guaranteeing global C^∞ regularity.
- **The Supercritical Barrier and Conditional Blowup.** Under the standard scaling ($\alpha_x = 1, \alpha_t = 2$), the three-dimensional NSE are energy-supercritical, a fact often cited as an obstruction to global regularity (see Tao's discussion of the "supercritical barrier" in [12]). By contrast, the symmetry-based embedding and a priori norm bounds derived here circumvent the supercriticality obstruction: the isobaric weight condition $\beta_x/\beta_t > 3/2$ ensures uniform control of energy and vorticity norms, which in turn rules out finite–time singularity.
- **The Importance of Self-Similar Solutions** A further landmark is the work of Giga and Miyakawa [5], who established global well–posedness of the NSE for small initial vorticity data in Morrey spaces, a framework broad enough to include measure–valued vorticity such as vortex filaments and rings. Their analysis identified forward self–similar solutions, showing that they can govern the large–time asymptotics of the flow. Their contribution represents one of the earliest rigorous demonstrations that self–similarity is not merely a heuristic symmetry, but a genuine mechanism for extending the range of initial data leading to global smooth solutions. The present work builds on this perspective: by embedding arbitrary smooth divergence–free initial data into globally defined self–similar families, we provide a complementary route to global regularity. Thus, both approaches underscore the role of self–similarity in extending classical results and offering new pathways toward the NSE regularity problem.
- **Relation to Functional-Analytic Frameworks.** Classical treatments of NSE regularity (e.g. [20], [21]) rely on functional analysis estimates and energy inequalities. While those methods provide the backbone of existence and uniqueness theory, the present approach complements

them by exploiting group-invariant structures to obtain global a priori estimates. In particular, embedding into a self-similar solution family and invoking classical well-posedness theorems suffices to identify a unique, globally smooth solution for any smooth initial data.

Together, these comparisons underscore the pertinence of the self-similar symmetry approach: it retains full compatibility with established PDE theory while extending global regularity beyond the small-data and energy-critical regimes that have traditionally constrained three-dimensional NSE analysis.

Appendix A. Detailed Proof of Theorem A.1

Theorem A.1. *Under standard conditions $\alpha_x = 1, \alpha_t = 2$, given a smooth, space-periodic, divergence-free, non-self-similar initial condition $\vec{u}_0(x, y, z)$, the incompressible NSE will always yield an infinitely differentiable, space-periodic, non-self-similar smooth solution $\vec{u}^*(x, y, z, t)$, such that $\vec{u}^*(x, y, z, 0) = \vec{u}_0(x, y, z)$. The solution $\vec{u}^*(x, y, z, t)$ is not subject to scaling-induced blowup for all time, thus ensuring bounded energy of the system in $(\mathbb{R}^3 \times [0, \infty))$. These desired properties are achieved by embedding the non-self-similar initial condition $\vec{u}_0$ into a self-similar space-periodic function $U\boldsymbol{\psi}$, which in turn results in the embedding of the non-self-similar smooth solution $\vec{u}^*(x, y, z, t)$ into a self-similar space-periodic function $U\boldsymbol{\varphi}$, thus preventing any rescalings of $\vec{u}^*(x, y, z, t)$. To yield the desired results, the self-similar functions $U\boldsymbol{\psi}, U\boldsymbol{\varphi}$ must have non-standard isobaric weight $\beta_x - \beta_t$ carried by the rescaleable parameter U, where $\beta_x/\beta_t > 3/2$, while the functions $\boldsymbol{\psi}, \boldsymbol{\varphi}$ have zero isobaric weight.*

Proof. All self-similar solutions (3.1) exist as mathematically valid solutions of the NSE. This includes the standard cases $\alpha_x = 1, \alpha_t = 2$, where the viscosity is the same at all scales: such solutions are given in (4.2) and (4.3). Here, we study only the mismatched solutions (5.1) described in Section 5, which can be summarized as

$$\vec{u} = U\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \tag{A.1}$$

where the isobaric weight of $\vec{u}$ is equal to $\beta_x - \beta_t$ and comes from the multiplying parameter while $\mathbf{F}$ has zero isobarity. Here L_1, L_2, L_3, T, U are nonzero parameters that scale the same way as x, y, z, t, u accordingly. NSE solutions of this form are not only known to exist (some examples given above), but are very common in physics; and one can then conclude that the NSE solutions (A.1) always exist.

We require the self-similar solutions (A.1) to have non-standard isobarity $\beta_x - \beta_t$ that does not match the standard conditions $\alpha_x = 1, \alpha_t = 2$; we also require $\beta_x/\beta_t > 3/2$, as per Lemma 6.2.

The space-periodic solutions

$$\vec{u} = U\boldsymbol{\varphi}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \tag{A.2}$$

are subset of the self-similar (A.1); moreover, space-periodic NSE solutions of the form (A.2) are known to exist (e.g. the solutions of the Taylor-Green vortex above), and from all the above one can then conclude that the NSE solutions (A.2) always exist. Note, that we have chosen (A.2) to be periodic in space only, and not in time.

At $t = 0$

$$U\boldsymbol{\varphi}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, 0\right) = U\boldsymbol{\psi}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}\right), \tag{A.3}$$

and all such space-periodic self-similar solutions with isobaric weight equal to $\beta_x - \beta_t$ (A.2) reduce to space-periodic self-similar functions of x, y, z and isobaric weight of $\beta_x - \beta_t$, where $\boldsymbol{\psi}$ need not be steady-state solutions of the NSE.

Suppose we are given a smooth, real-valued, periodic and divergence-free vector field $\vec{u}_0$, defined on a rectangular domain $[0, L_1] \times [0, L_2] \times [0, L_3]$, where L_i are constants. Generally speaking, space-periodic functions are not self-similar. It can always be expanded in a real Fourier series as:

$$\vec{u}_0\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3}\right) = U \sum_{n_1,n_2,n_3\in\mathbb{Z}_+^3} \left[\mathbf{C}_{n_1,n_2,n_3} \cos\left(\frac{2\pi n_1 x}{L_1}+\frac{2\pi n_2 y}{L_2}+\frac{2\pi n_3 z}{L_3}\right) \right.$$

$$\left. + \mathbf{S}_{n_1,n_2,n_3} \sin\left(\frac{2\pi n_1 x}{L_1}+\frac{2\pi n_2 y}{L_2}+\frac{2\pi n_3 z}{L_3}\right) \right], \tag{A.4}$$

where:

- U is a constant,
- $\mathbf{C}_{n_1,n_2,n_3}, \mathbf{S}_{n_1,n_2,n_3} \in \mathbb{R}^3$ are real-valued vector coefficients,
- The sum is taken over all non-negative integers $n_1, n_2, n_3 \in \mathbb{Z}_+$,
- The terms $\frac{2\pi n_1}{L_1}$, $\frac{2\pi n_2}{L_2}$, and $\frac{2\pi n_3}{L_3}$ are the spatial frequencies in the x, y, and z directions, respectively.

Therefore, the form of the ansatz $\vec{u}_0$ is justified. Now we use Lemma 5.1- we construct a self-similar function $\boldsymbol{\psi}$ with mathematical form identical to $\vec{u}_0$, with the exception that L_1, L_2, L_3, U are no longer constants, but parameters that rescale as x, y, z, u accordingly:

$$U\boldsymbol{\psi}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3}\right).$$

It is seen, that as long as $\vec{u}_0$ is divergence-free, so is $U\boldsymbol{\psi}$. It is also seen, that $\vec{u}_0 \in \boldsymbol{\psi}$, that is $\vec{u}_0$ is embedded in the self-similar function $\boldsymbol{\psi}$ at the identity scale, so that

$$\vec{u}_0 = U\boldsymbol{\psi}|_{k=1}. \tag{A.5}$$

The original initial condition $\vec{u}_0$ is reproduced at scale $k = 1$ by rescaling the parameter U. Notice that $U\boldsymbol{\psi}$ is a self-similar space-periodic function, or a family, a set of functions corresponding to various rescalings of U; while $\vec{u}_0$ is only one space-periodic function of the aforementioned family, achieved at scale $k = 1$.

It is well known (from the Fujita-Kato Theorem 7.1) that, given a smooth, space-periodic, divergence free, self-similar initial condition $U\boldsymbol{\psi}$, one can always construct an unique, infinitely differentiable, space-periodic, self-similar NSE solution

$$U\boldsymbol{\varphi}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right) \tag{A.6}$$

such that

$$U\boldsymbol{\varphi}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},0\right) = U\boldsymbol{\psi}\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3}\right),$$

which exists in the time interval $[0,\tau)$ [1], [17] - [21]. We see that, the space-periodic solution $U\boldsymbol{\varphi}$ of Eq.(A.6), being a self-similar solution with isobarity equal to $\beta_x - \beta_t$, belongs to the set of self-similar solutions of Eq. (A.2).

Analogously, it is well known that, given a smooth, space-periodic, divergence free, non-self-similar initial condition $\vec{u}_0$, one can always construct an unique, infinitely differentiable, space-periodic, non-self-similar NSE solution

$$U\vec{u}^*\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},\frac{t}{T}\right) \tag{A.7}$$

such that

$$U\vec{u}^*\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},0\right) = \vec{u}_0\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3}\right),$$

which exists in the time interval $[0,\tau)$ [1], [17] - [21]. In this case, L_1, L_2, L_3, T, U are fixed, non-rescaleable constants. Analogously with (A.5), it follows that $U\vec{u}^* \in U\boldsymbol{\varphi}$, that is, $U\vec{u}^*$ is embedded in the self-similar solution $U\boldsymbol{\varphi}$ so that

$$U\vec{u}^* = U\boldsymbol{\varphi}|_{k=1}.$$

The original solution $U\vec{u}^*$ corresponds to the original initial condition $\vec{u}_0$, such that $U\vec{u}^*(\cdot, 0) = \vec{u}_0$ and is reproduced at scale $k = 1$ by rescaling of the parameter U of the self-similar function $U\boldsymbol{\varphi}$. Notice that $U\boldsymbol{\varphi}$ is a self-similar solution, or a family, a set of solutions corresponding to different rescalings of U; while $U\vec{u}^*$ is only one solution of the aforementioned family, achieved at scale $k = 1$.

The self-similar solutions (A.1) have isobaric weight $\beta_x - \beta_t$ with $\beta_x/\beta_t > 3/2$; but rescaled under standard conditions $\alpha_x = 1, \alpha_t = 2$ remain smooth for all time according to the Beale-Kato-Majda criterion [15], as shown in Lemma 6.2. The self-similar solutions (A.1) do not exhibit scaling-induced blow-up of the vorticity, the energy or the velocity. It follows, that the energy of such solutions will remain bound for all time.

Because $U\vec{u}^*$ (corresponding to the original initial condition $\vec{u}_0$) is embedded in the self-similar solution $U\boldsymbol{\varphi}$ at scale $k = 1$, and both are solutions to the NSE with the same initial data, uniqueness implies that $U\vec{u}^* = U\boldsymbol{\varphi}|_{k=1}$ for all time. Since $U\boldsymbol{\varphi}$ is globally smooth, it follows that $U\vec{u}^*$ remains smooth throughout $\mathbb{R}^3 \times [0, \infty)$.

Regarding the physics of the self-similar solutions (A.1): smoothness, or infinite differentiability is easily shown to be a property of the self-similar solutions (A.1) (see Theorem 5.3). Except for the multiplying constant, they are isobaric polynomials of zero weight, or can be Taylor expansions of closed-form functions which depend on zero-weight arguments, e.g. $x/L, t/T$; or are the ratio of such functions. Since the energy of the system comes solely from the smooth, space-periodic initial velocity $\vec{u}_0$ (and pressure), the initial energy is bound; (A.1) cannot grow at infinity. This is always possible in a zero-weight function (note that the isobarity comes solely through U), since decay can always be ensured through the presence of exponential functions, e.g. $\exp[-(x^2 + y^2 + z^2)/L^2]$. Smoothness and decay in t is also required, since the system energy must be conserved. This also is always possible in a zero-weight function, since decay can always be ensured through the presence of exponential functions, e.g. $\exp[-t/T]$. The Theorem 7.1 of Fujita-Kato guarantees smoothness of the solution on $[0, \tau)$, where we have extended τ by ruling out blowup.

We can therefore conclude, that given any arbitrary, smooth, divergence free, space-periodic function (A.4) in the initial moment $t = 0$, the incompressible NSE always has a self-similar solution with isobarity of $\beta_x - \beta_t$ with $\beta_x/\beta_t > 3/2$ (A.2), which is infinitely differentiable, being a constant U multiplying a zero-weight isobaric function; non-increasing in space due to the requirement of bound initial energy at $t = 0$ and non-increasing in time within the interval $[0, \tau)$ due to the Fujita-Kato Theorem 7.1. The latter guarantees that system energy will be bound in $[0, \tau)$. To maintain these properties for all time under infinite rescaling while we are modeling the fluid as a continuum, rescaling of the environment x, y, z, t must be performed according to standard conditions $\alpha_x = 1, \alpha_t = 2$. For all time, the original solution $U\vec{u}^*$ corresponding to the original initial condition $\vec{u}_0$, will remain embedded into the self-similar solution $U\boldsymbol{\varphi}$ at scale $k = 1$; this embedding prevents any rescalings of $U\vec{u}^*$, while the rescalings of the self-similar solution $U\boldsymbol{\varphi}$ always keep the vorticity norm bound by virtue of the mismatch between the isobaric weight $\beta_x - \beta_t$ of the self-similar solution and the standard environment $\alpha_x = 1, \alpha_t = 2$, as shown in Lemma 6.2. □

Appendix B. Detailed Proof of Theorem B.1 and Corollary B.2

Theorem B.1. *Under standard conditions $\alpha_x = 1, \alpha_t = 2$, given a smooth, non-self-similar, Schwartz-class, divergence-free initial condition $\vec{u}_0(x, y, z)$, the incompressible NSE will always yield an infinitely differentiable, non-self-similar, Schwartz-class, smooth solution $\vec{u}^*(x, y, z, t)$, such that $\vec{u}^*(x, y, z, 0) = \vec{u}_0(x, y, z)$. The solution $\vec{u}^*(x, y, z, t)$ is not subject to scaling-induced blowup for all time, thus ensuring bounded energy of the system in $(\mathbb{R}^3 \times [0, \infty))$. These desired properties are achieved by embedding the non-self-similar initial condition $\vec{u}_0$ into a self-similar function $U\boldsymbol{s}$, which in turn results in the embedding of the non-self-simialr smooth solution $\vec{u}^*(x, y, z, t)$ into a self-similar function $U\boldsymbol{S}$, thus preventing any rescalings of $\vec{u}^*(x, y, z, t)$. To yield the desired results, the self-similar functions $U\boldsymbol{s}, U\boldsymbol{S}$ must have non-standard isobaric weight $\beta_x - \beta_t$, carried by the rescaleable parameter U where $\beta_x/\beta_t > 3/2$, while the functions $\boldsymbol{s}, \boldsymbol{S}$ have zero isobaric weight.*

Proof. In this section, we follow closely the proof given in the previous section. Here, we study only Bouton's mismatched self-similar solutions 5.1 with isobaric weight $\beta_x - \beta_t$, summarized as (A.1) under standard conditions $\alpha_x = 1, \alpha_t = 2$. They are continuous at $t = 0$; their isobaric weight comes from the multiplying parameter while $\mathbf{F}$ has zero isobarity. L_1, L_2, L_3, T, U are nonzero parameters that scale the same way as x, y, z, t, u accordingly.

The solutions

$$\vec{u} = U\mathbf{S}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \tag{B.1}$$

are a subset of (A.1); the self-similar function $\mathbf{S}$ has zero isobaric weight by virtue of the scaling parameters L_1, L_2, L_3, T.

At $t = 0$

$$U\mathbf{S}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, 0\right) = U\mathbf{s}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}\right), \tag{B.2}$$

and all such self-similar solutions with isobaric weight equal to $\beta_x - \beta_t$ (B.1) reduce to self-similar functions of x, y, z and isobaric weight of $\beta_x - \beta_t$, where $\mathbf{s}$ need not be steady-state solutions of the NSE.

Schwartz-class functions are generally non-self-similar and have Gaussian peak form $\exp(-r)$, a polynomial multiplying a Gaussian $P(x, y, z)\exp(-r)$, a bump function form or a convolution of the above functions. Suppose we are given a smooth, real-valued and divergence-free vector field $\vec{u}_0$ of Schwartz-class, defined with the fixed constant scale factors L_1, L_2, L_3

$$\vec{u}_0\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}\right) = U\mathbf{s}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}\right), \tag{B.3}$$

where U is a fixed constant.

Now we use Lemma 5.1- we construct a self-similar function $\mathbf{s}$ with mathematical form identical to $\vec{u}_0$, with the exception that L_1, L_2, L_3, U are no longer constants, but parameters that rescale as x, y, z, u accordingly:

$$U\mathbf{s}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}\right).$$

It is seen, that as long as $\vec{u}_0$ is divergence-free, so is $U\mathbf{s}$. It is also seen, that $\vec{u}_0 \in \mathbf{s}$, that is $\vec{u}_0$ is embedded in the self-similar function $\mathbf{s}$, so that

$$\vec{u}_0 = U\mathbf{s}|_{k=1}. \tag{B.4}$$

The original initial condition $\vec{u}_0$ is reproduced at scale $k = 1$ by rescaling the parameter U. Notice that $U\mathbf{s}$ is a self-similar function, or a family, a set of functions corresponding to different rescalings of U; while $\vec{u}_0$ is only one Schwartz-class function of the aforementioned family, achieved at scale $k = 1$.

It is well known (from the Fujita-Kato Theorem 7.1) that, given a smooth, divergence free, self-similar initial condition $U\mathbf{s}$, one can always construct an unique, infinitely differentiable, self-similar NSE solution

$$U\mathbf{S}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right) \tag{B.5}$$

such that

$$U\mathbf{S}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, 0\right) = U\mathbf{s}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}\right),$$

which exists in the time interval $[0, \tau)$ [1], [17] - [21]. We see that, the solution $U\mathbf{S}$ of Eq.(B.5), being a self-similar solution with isobarity equal to $\beta_x - \beta_t$, belongs to the set of self-similar solutions Eq. (B.1).

Analogously, it is well known that, given a smooth, Schwartz-class, divergence free, non-self-similar initial condition $\vec{u}_0$, one can always construct an unique, infinitely differentiable, Schwartz-class, non-self-similar NSE solution

$$U\vec{u}^*\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right) \tag{B.6}$$

such that

$$U\vec{u}^*\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3},0\right)=\vec{u}_0\left(\frac{x}{L_1},\frac{y}{L_2},\frac{z}{L_3}\right),$$

which exists in the time interval $[0,\tau)$ [1], [17] - [21]. In this case, L_1, L_2, L_3, T, U are fixed, non-rescaleable constants. Analogously with (B.4), it follows that $U\vec{u}^* \in U\mathbf{S}$, that is, $U\vec{u}^*$ is embedded in the self-similar solution $U\mathbf{S}$ so that

$$U\vec{u}^* = U\mathbf{S}|_{k=1}.$$

The original solution $U\vec{u}^*$ corresponds to the original initial condition $\vec{u}_0$, such that $U\vec{u}^*(\cdot,0)=\vec{u}_0$ and is reproduced at scale $k=1$ by rescaling of the parameter U of the self-similar function $U\mathbf{S}$. Notice that $U\mathbf{S}$ is a self-similar solution, or a family, a set of solutions corresponding to different rescalings of U; while $U\vec{u}^*$ is only one solution of the aforementioned family, achieved at scale $k=1$.

Analogously with the reasoning in Theorem A.1, we limit our study only to self-similar solutions with isobarity $\beta_x-\beta_t$ with $\beta_x/\beta_t > 3/2$ and we argue that by virtue of Lemma 6.2, the self-similar solution remains smooth for all time when rescaled under standard conditions $\alpha_x=1, \alpha_t=2$ according to the Beale-Kato-Majda criterion [15], since scaling-indiced blowup is impossible in this case.

Because $U\vec{u}^*$ (corresponding to the original Schwartz-class initial condition $\vec{u}_0$) is embedded in the self-similar solution $U\mathbf{S}$ at scale $k=1$, it will possess the characteristics of $U\mathbf{S}$ and will thus remain smooth throughout $(\mathbb{R}^3\times[0,\infty))$, because $U\mathbf{S}$ remains smooth.

We can be confident in the smoothness of $U\mathbf{S}$, as well as in its bound energy for all time through arguments, analogous to those given at the end of the proof of Theorem A.1.

□

Corollary B.2. *Under standard scaling conditions $\alpha_x=1$, $\alpha_t=2$, let*

$$u_0(x,y,z)$$

be any divergence-free, initial datum on $\mathbb{R}^3$, which satisfies the prerequisites of the Fujita-Kato Theorem 7.1. Then the incompressible NSE admit a unique, smooth solution

$$u^*(x,y,z,t) \;\in\; C^\infty\big(\mathbb{R}^3\times[0,\tau)\big).$$

with

$$u^*(x,y,z,0)=u_0(x,y,z)\,.$$

Moreover u^* can be embedded as the $k=1$ member of a self-similar family of the form (5.1)

$$T^{(\beta_x-\beta_t)/\beta_t}\,F\Big(\tfrac{x}{L_1},\tfrac{y}{L_2},\tfrac{z}{L_3},\tfrac{t}{T}\Big),$$

where $\beta_x/\beta_t>3/2$. Lemma 6.2 then shows that under the standard scaling no velocity, energy, or vorticity norm grows with k, and the Beale–Kato–Majda criterion rules out finite-time blowup. Hence

$$u^*(x,y,z,t) \;\in\; C^\infty\big(\mathbb{R}^3\times[0,\infty)\big)$$

with uniformly bounded energy on $\mathbb{R}^3\times[0,\infty)$.

Proof. Let $u_0(x,y,z)$ be any divergence-free vector field on $\mathbb{R}^3$, which satisfies the prerequisites of the Fujita-Kato Theorem 7.1. By Lemma 5.1 and Corollary 5.2, u_0 embeds as the $k=1$ member of a one-parameter self-similar family (5.2)

$$T^{(\beta_x-\beta_t)/\beta_t}\,F\Big(\tfrac{x}{L_1},\tfrac{y}{L_2},\tfrac{z}{L_3},0\Big).$$

Classical existence and uniqueness theorems (e.g. Fujita-Kato [1], [17] - [21]) then guarantee that there is a unique, smooth solution (5.1)

$$T^{(\beta_x-\beta_t)/\beta_t}\,F\Big(\tfrac{x}{L_1},\tfrac{y}{L_2},\tfrac{z}{L_3},\tfrac{t}{T}\Big),$$

evolving from u_0 on an interval $[0,\tau)$ and having u^* embedded as the $k=1$ member of the self-similar family (5.1). Lemma 6.2 provides uniform bounds on the vorticity and energy norms under the standard scaling ($\alpha_x = 1, \alpha_t = 2$), and the Beale–Kato–Majda criterion [15] ensures no finite-time singularity can occur. $\square$

Appendix C. Examples of Solutions in the $UF\left(\frac{x}{L}, \frac{t}{T}\right)$ Form

To further illustrate the general structure of the self-similar solutions developed in this work, we present explicit examples of solutions of the form

$$u(x, y, z, t) = UF\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right),$$

where F is a smooth function depending on dimensionless spatial and temporal variables. These examples demonstrate the smoothness, boundedness, and structural stability of solutions arising from both periodic and Schwartz-class initial data.

We provide two worked examples illustrating explicit solutions of the above form, where U is a constant (velocity scale), L_i are length scales, T is a time scale, and F is a smooth function of the dimensionless variables $(x_i/L_i, t/T)$.

These examples demonstrate explicit, smooth, non-blowing-up solutions consistent with the structure used throughout the article.

C.1. **Example: Periodic, Smooth Initial Data.**

Let $L > 0$ be a length scale and set

$$\xi_1 = \frac{x}{L}, \qquad \xi_2 = \frac{y}{L}, \qquad \xi_3 = \frac{z}{L}.$$

Define a (scaled) ABC / Beltrami field V by

$$V\Big(\tfrac{x}{L}, \tfrac{y}{L}, \tfrac{z}{L}\Big) = \begin{pmatrix} \sin\big(\frac{z}{L}\big) + \cos\big(\frac{y}{L}\big) \\ \sin\big(\frac{x}{L}\big) + \cos\big(\frac{z}{L}\big) \\ \sin\big(\frac{y}{L}\big) + \cos\big(\frac{x}{L}\big) \end{pmatrix}.$$

This field is a known example of a fluid flow with chaotic trajectories [25]. It is $2\pi L$–periodic in each of x, y, z, smooth, and divergence-free. In the ξ-variables it satisfies

$$\nabla_\xi \times V(\xi) = V(\xi), \qquad \Delta_\xi V(\xi) = -V(\xi).$$

Thus under the spatial rescaling $x \mapsto \xi = x/L$ we have

$$\nabla_x \times V\Big(\tfrac{x}{L}\Big) = \frac{1}{L} V\Big(\tfrac{x}{L}\Big), \qquad \Delta_x V\Big(\tfrac{x}{L}\Big) = -\frac{1}{L^2} V\Big(\tfrac{x}{L}\Big).$$

Let the time-dependent amplitude be

$$a(t) = U \exp\Big(-\nu \frac{t}{L^2}\Big),$$

where $U = a(0)$ is the initial amplitude and $\nu > 0$ is the viscosity. Define the velocity field in full coordinates by

$$u(x, y, z, t) = a(t) \begin{pmatrix} \sin\big(\frac{z}{L}\big) + \cos\big(\frac{y}{L}\big) \\ \sin\big(\frac{x}{L}\big) + \cos\big(\frac{z}{L}\big) \\ \sin\big(\frac{y}{L}\big) + \cos\big(\frac{x}{L}\big) \end{pmatrix}.$$

Verification. Using the Beltrami identity in ξ-variables,

$$(V\cdot\nabla_\xi)V = \nabla_\xi\Big(\tfrac12|V|^2\Big) - V\times(\nabla_\xi\times V) = \nabla_\xi\Big(\tfrac12|V|^2\Big),$$

so

$$(u\cdot\nabla_x)u = a(t)^2\,\nabla_x\Big(\tfrac12|V|^2\Big(\tfrac{x}{L},\tfrac{y}{L},\tfrac{z}{L}\Big)\Big),$$

a pure gradient. The viscous term is

$$\nu\Delta_x u = -\frac{\nu}{L^2}u(x,y,z,t).$$

Thus the Navier–Stokes momentum equation reduces to the scalar ODE

$$a'(t) = -\frac{\nu}{L^2}a(t),$$

solved by the chosen amplitude $a(t)$. The pressure balances the gradient term:

$$p(x,y,z,t) = \frac{a(t)^2}{2}\,|V|^2\Big(\tfrac{x}{L},\tfrac{y}{L},\tfrac{z}{L}\Big) + q(t),$$

with $q(t)$ arbitrary (e.g. $q\equiv 0$).

Explicit solution. The NSE admits the smooth, global solution

$$\boxed{\begin{aligned} u(x,y,z,t) &= U\exp\Big(-\nu\frac{t}{L^2}\Big)\begin{pmatrix}\sin(\frac{z}{L})+\cos(\frac{y}{L})\\ \sin(\frac{x}{L})+\cos(\frac{z}{L})\\ \sin(\frac{y}{L})+\cos(\frac{x}{L})\end{pmatrix},\\ p(x,y,z,t) &= \frac{U^2}{2}\exp\Big(-2\nu\frac{t}{L^2}\Big)\left|\begin{pmatrix}\sin(\frac{z}{L})+\cos(\frac{y}{L})\\ \sin(\frac{x}{L})+\cos(\frac{z}{L})\\ \sin(\frac{y}{L})+\cos(\frac{x}{L})\end{pmatrix}\right|^2 + q(t).\end{aligned}}$$

Properties.

- $u(\cdot,t)$ is smooth and $2\pi L$–periodic in (x,y,z) for each t.
- $\nabla\cdot u = 0$ for all t.
- The solution exists globally in time, decays exponentially as $t\to\infty$, and no singularity arises.

C.2. **Example: Schwartz-Class Initial Data.** Let $U,L,T,\nu>0$ be rescaleable parameters. Define the self-similar variables

$$\xi := \frac{kx}{L},\qquad \tau := \frac{k^2t}{T},\qquad r^2 := x_1^2+x_2^2+x_3^2.$$

1. Self-similar profile. Introduce the rotation vector

$$e(\xi) := \begin{pmatrix}-\xi_2\\ \xi_1\\ 0\end{pmatrix},$$

and define the smooth, divergence-free Gaussian profile

$$F(\xi,\tau) := e^{-\tau}e^{-|\xi|^2}e(\xi),\qquad \nabla_\xi\cdot F = 0.$$

2. Embedding in a one-parameter family. Define the one-parameter self-similar family

$$u_k(x,t) := U\,F\Big(\frac{kx}{L},k^2\frac{t}{T}\Big),\qquad k>0.$$

3. Initial data ($k = 1$). Fix $k = 1$ to obtain the initial velocity:

$$u(x,t) = u_1(x,t) = U\,F\Big(\frac{x}{L}, \frac{t}{T}\Big) = Ue^{-t/T}e^{-|x|^2/L^2}\begin{pmatrix}-x_2\\ x_1\\ 0\end{pmatrix}.$$

At $t = 0$, this gives

$$u(x,0) = Ue^{-|x|^2/L^2}\begin{pmatrix}-x_2\\ x_1\\ 0\end{pmatrix}.$$

4. Forcing term. Compute

$$S(r) := \frac{1}{T} - \frac{10\nu}{L^2} + \frac{4\nu r^2}{L^4}.$$

Then the forcing $f(x,t) = \partial_t u + (u\cdot\nabla)u - \nu\Delta u$ can be written as:
Vector form.

$$\boxed{f(x,t) = -U\,e^{-t/T}e^{-|x|^2/L^2}S(r)\begin{pmatrix}-x_2\\ x_1\\ 0\end{pmatrix} - U^2e^{-2t/T}e^{-2|x|^2/L^2}\begin{pmatrix}x_1\\ x_2\\ 0\end{pmatrix}.}$$

Component form.

$$\begin{aligned}
f_1(x,t) &= Ue^{-t/T}e^{-|x|^2/L^2}\,x_2\,S(r) - U^2e^{-2t/T}e^{-2|x|^2/L^2}x_1,\\
f_2(x,t) &= -Ue^{-t/T}e^{-|x|^2/L^2}\,x_1\,S(r) - U^2e^{-2t/T}e^{-2|x|^2/L^2}x_2,\\
f_3(x,t) &= 0.
\end{aligned}$$

Remarks.

- The self-similar embedding shows the one-parameter family and how $k = 1$ fixes the initial data.
- Each term in f is a polynomial in the coordinates times a Gaussian, so $f(\cdot,t)$ is Schwartz-class for all $t \geq 0$.
- With this forcing, (u,p) solves the *forced* incompressible NSE:

$$\partial_t u + (u\cdot\nabla)u = \nu\Delta u + f, \qquad \nabla\cdot u = 0.$$

- The pressure p is obtained (up to an additive function of time) from $\Delta p = -\nabla\cdot((u\cdot\nabla)u) + \nabla\cdot(\nu\Delta u - \partial_t u)$; it is smooth and rapidly decaying.

Appendix D. Example: Embedding a Periodic Initial Condition into a Self-Similar Function

D.1. **Initial Condition.** We consider a smooth, divergence-free, space-periodic velocity field:

$$\vec{u}_0(x,y) = \begin{bmatrix}\sin(x)\cos(y)\\ -\cos(x)\sin(y)\end{bmatrix}$$

This function is:

- Smooth (C^∞)
- Periodic in x and y with period 2π
- Divergence-free: $\nabla\cdot\vec{u}_0 = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

D.2. **Self-Similar Embedding.** We embed $\vec{u}_0$ into a self-similar function of the form:

$$\vec{u}(x,y,t) = T^{\frac{\beta_x-\beta_t}{\beta_t}} \cdot \vec{F}\left(\frac{x}{L}, \frac{y}{L}, \frac{t}{T}\right)$$

Let us choose:

$$\vec{F}\left(\frac{x}{L}, \frac{y}{L}, \frac{t}{T}\right) = \begin{bmatrix} \sin\left(\frac{x}{L}\right)\cos\left(\frac{y}{L}\right) \\ -\cos\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right) \end{bmatrix}$$

This gives the full self-similar velocity field:

$$\vec{u}(x,y,t) = T^{\frac{\beta_x-\beta_t}{\beta_t}} \begin{bmatrix} \sin\left(\frac{x}{L}\right)\cos\left(\frac{y}{L}\right) \\ -\cos\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right) \end{bmatrix}$$

We choose the isobaric weight such that:

$$\frac{\beta_x}{\beta_t} > \frac{3}{2} \quad \text{(to avoid blowup under standard conditions)}$$

For instance, let $\beta_x = 5$, $\beta_t = 3$, so:

$$T^{\frac{\beta_x-\beta_t}{\beta_t}} = T^{\frac{2}{3}}$$

Therefore, the final function is:

$$\vec{u}(x,y,t) = T^{2/3} \begin{bmatrix} \sin\left(\frac{x}{L}\right)\cos\left(\frac{y}{L}\right) \\ -\cos\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right) \end{bmatrix}$$

At $t = 0$, setting $T = 1$, $L = 1$ recovers the original initial condition:

$$\vec{u}(x,y,0) = \vec{u}_0(x,y)$$

D.3. **Vorticity.** The vorticity in 2D is defined as:

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Compute derivatives:

$$\frac{\partial v}{\partial x} = \frac{T^{2/3}}{L}\sin\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right), \quad \frac{\partial u}{\partial y} = -\frac{T^{2/3}}{L}\sin\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right)$$

Therefore:

$$\omega(x,y,t) = \frac{2T^{2/3}}{L}\sin\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right)$$

Under standard conditions $\alpha_x = 1, \alpha_t = 2$, the vorticity factor $T^{2/3}/L$ rescales as

$$\left(\frac{T^{2/3}}{L}\right)' = \left(\frac{(k^{\alpha_t}T)^{2/3}}{k^{\alpha_x}L}\right) = \left(\frac{k^{4/3}T}{kL}\right) = \frac{k^{1/3}T}{L},$$

and the rest is an isobaric polynomial of weight zero (does not rescale).

D.4. **Conclusion.** We have embedded a smooth, space-periodic initial condition into a self-similar function with isobaric weight satisfying $\beta_x/\beta_t > 3/2$. The resulting velocity and vorticity remain smooth for all time under standard rescaling.

Appendix E. Example: Embedding a Smooth Schwartz-Class Initial Condition

1. Smooth Schwartz-Class Initial Condition. Let us define the following 3D velocity field as initial data:

$$\vec{u}_0(x,y,z) = \begin{bmatrix} yze^{-r^2} \\ xze^{-r^2} \\ -2xye^{-r^2} \end{bmatrix}, \quad \text{where } r^2 = x^2+y^2+z^2$$

This vector field satisfies:

- **Smoothness**: All components are C^∞.
- **Rapid decay at infinity**: Gaussian term e^{-r^2} ensures Schwartz-class behavior.
- **Divergence-free**:

$$\nabla \cdot \vec{u}_0 = \partial_x(yze^{-r^2}) + \partial_y(xze^{-r^2}) + \partial_z(-2xye^{-r^2}) = 0$$

2. Self-Similar Embedding. Let $\vec{u}_0$ be embedded in a self-similar function of the form:

$$\vec{u}(x,y,z,t) = T^{\frac{\beta_x-\beta_t}{\beta_t}} \vec{F}\left(\frac{x}{L}, \frac{y}{L}, \frac{z}{L}, \frac{t}{T}\right)$$

Choose:

$$\vec{F}\left(\frac{x}{L}, \frac{y}{L}, \frac{z}{L}, \frac{t}{T}\right) = \begin{bmatrix} \frac{yz}{L^2} \cdot e^{-\frac{x^2+y^2+z^2}{L^2}} \\ \frac{xz}{L^2} \cdot e^{-\frac{x^2+y^2+z^2}{L^2}} \\ \frac{-2xy}{L^2} \cdot e^{-\frac{x^2+y^2+z^2}{L^2}} \end{bmatrix}$$

Thus, the self-similar velocity field is:

$$\vec{u}(x,y,z,t) = T^{\frac{\beta_x-\beta_t}{\beta_t}} \begin{bmatrix} \frac{yz}{L^2} e^{-\frac{x^2+y^2+z^2}{L^2}} \\ \frac{xz}{L^2} e^{-\frac{x^2+y^2+z^2}{L^2}} \\ \frac{-2xy}{L^2} e^{-\frac{x^2+y^2+z^2}{L^2}} \end{bmatrix}$$

we aim to compute the vorticity

$$\vec{\omega} = \nabla \times \vec{u}.$$

Let us denote the exponential factor as

$$f(x,y,z) = e^{-r^2/L^2}, \quad \text{and set } C = \frac{T^{(\beta_x-\beta_t)/\beta_t}}{L^2}.$$

Then

$$\vec{u}(x,y,z,t) = C \cdot f(x,y,z) \begin{bmatrix} yz \\ xz \\ -2xy \end{bmatrix}.$$

First component: $\omega_1 = \partial_y u_z - \partial_z u_y$.

$$\partial_y u_z = \partial_y\left(-2xy \cdot Cf\right) = -2xC\left(f + y \cdot \partial_y f\right) = -2xCf\left(1 - \frac{2y^2}{L^2}\right),$$

$$\partial_z u_y = \partial_z\left(xz \cdot Cf\right) = xC\left(f + z \cdot \partial_z f\right) = xCf\left(1 - \frac{2z^2}{L^2}\right),$$

$$\begin{aligned} \Rightarrow \quad \omega_1 = \partial_y u_z - \partial_z u_y &= -Cfx\left[2\left(1 - \frac{2y^2}{L^2}\right) + \left(1 - \frac{2z^2}{L^2}\right)\right] \\ &= -Cfx\left(3 - \frac{4y^2+2z^2}{L^2}\right). \end{aligned}$$

Second component: $\omega_2 = \partial_z u_x - \partial_x u_z$.

$$\partial_z u_x = \partial_z \left(yz \cdot Cf\right) = yCf\left(1 - \frac{2z^2}{L^2}\right),$$

$$\partial_x u_z = \partial_x \left(-2xy \cdot Cf\right) = -2yCf\left(1 - \frac{2x^2}{L^2}\right),$$

$$\Rightarrow \quad \omega_2 = \partial_z u_x - \partial_x u_z = Cfy\left[1 - \frac{2z^2}{L^2} + 2\left(1 - \frac{2x^2}{L^2}\right)\right]$$
$$= Cfy\left(3 - \frac{2z^2 + 4x^2}{L^2}\right).$$

Third component: $\omega_3 = \partial_x u_y - \partial_y u_x$.

$$\partial_x u_y = \partial_x(xz \cdot Cf) = zCf\left(1 - \frac{2x^2}{L^2}\right),$$

$$\partial_y u_x = \partial_y(yz \cdot Cf) = zCf\left(1 - \frac{2y^2}{L^2}\right),$$

$$\Rightarrow \quad \omega_3 = zCf\left(\frac{2y^2 - 2x^2}{L^2}\right) = \frac{2zCf}{L^2}(y^2 - x^2).$$

Final result. The full vorticity vector is:

$$\vec{\omega}(x,y,z,t) = T^{\frac{\beta_x - \beta_t}{\beta_t}} \cdot \frac{e^{-r^2/L^2}}{L^2}\begin{bmatrix} -x\left(3 - \dfrac{4y^2 + 2z^2}{L^2}\right) \\ y\left(3 - \dfrac{4x^2 + 2z^2}{L^2}\right) \\ \dfrac{2z}{L^2}(y^2 - x^2)\end{bmatrix}.$$

Choose isobaric weights: $\beta_x = 5, \beta_t = 3 \Rightarrow \frac{\beta_x - \beta_t}{\beta_t} = \frac{2}{3}$
Then:

$$\vec{u}(x,y,z,t) = T^{2/3}\begin{bmatrix} \frac{yz}{L^2} e^{-\frac{x^2+y^2+z^2}{L^2}} \\ \frac{xz}{L^2} e^{-\frac{x^2+y^2+z^2}{L^2}} \\ \frac{-2xy}{L^2} e^{-\frac{x^2+y^2+z^2}{L^2}}\end{bmatrix}$$

At $t = 0$, set $T = 1, L = 1 \Rightarrow \vec{u}(x,y,z,0) = \vec{u}_0(x,y,z)$, so $\vec{u}_0 \in \vec{u}$

3. Vorticity Scaling. The vorticity is:

$$\vec{\omega}(x,y,z,t) = T^{\frac{2}{3}} \cdot \frac{e^{-r^2/L^2}}{L}\begin{bmatrix} -(x/L)\left(3 - \dfrac{4y^2 + 2z^2}{L^2}\right) \\ (y/L)\left(3 - \dfrac{4x^2 + 2z^2}{L^2}\right) \\ \dfrac{2z}{L^3}(y^2 - x^2)\end{bmatrix},$$

where again, as in the previous example the vorticity factor is $T^{2/3}/L$ which does not grow at small scales; and the rest is an isobaric finction of weight zero, which does not rescale. Thus, $\vec{\omega}(x,y,z,t)$ and the vorticity are smooth, rapidly decaying (Schwartz), and bounded for all t.

4. Conclusion. We have constructed a 3D smooth, divergence-free, Schwartz-class initial condition and embedded it into a self-similar function. The velocity and vorticity fields are all smooth for all time, and the solution avoids blowup by satisfying the isobarity condition $\beta_x/\beta_t > 3/2$ as required in the article.

Appendix F. Example of Isobaric Weight and Norm Behavior

To visualize the crucial role of the isobaric weight ratio $\beta_x/\beta_t > 3/2$ in preventing scaling-induced blowup, consider a simple self-similar velocity field of the form

$$\vec{u} = T^{\frac{\beta_x-\beta_t}{\beta_t}}\mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right)$$

where F is a fixed smooth profile (e.g. a Gaussian). Under the standard scaling $\alpha_x = 1, \alpha_t = 2$, one computes:

$$\|\omega_k\| \sim k^{\frac{2\beta_x-3\beta_t}{\beta_t}}, \quad \|u_k\| \sim k^{\frac{2(\beta_x-\beta_t)}{\beta_t}}, \quad E_k \ = \ \|u_k\| \sim k^{\frac{4\beta_x-\beta_t}{\beta_t}},$$

see Lemma 6.2.

Table 1: Exponents vs. β_x/β_t

$r = \dfrac{\beta_x}{\beta_t}$	ω-exp $= 2r - 3$	u-exp $= 2(r-1)$	E-exp $= 4r - 1$
−2.0	−7.0	−6.0	−9.0
−1.0	−5.0	−4.0	−5.0
−0.5	−4.0	−3.0	−3.0
0.0	−3.0	−2.0	−1.0
0.5	−2.0	−1.0	1.0
1.0	−1.0	0.0	3.0
1.2	−0.6	0.4	3.8
1.5	0.0	1.0	5.0
2.0	1.0	2.0	7.0
3.0	3.0	4.0	11.0

When $\beta_x/\beta_t > 3/2$, the vorticity exponent $2\beta_x - 3\beta_t > 0$ indicates decrease at fine scales under standard conditions, irrespective of the signs of β_x, β_t. In such regimes the combined energy and vorticity control, together with the Beale–Kato–Majda criterion, guarantees no finite-time blowup. This simple numerical illustration clarifies why the threshold $\beta_x/\beta_t = 3/2$ is pivotal in the scaling analysis.

Appendix G. Application of Bouton's Method to the Incompressible Euler Equations

G.1. **Scaling Transformation for the Euler Equations.** The incompressible Euler equations in three dimensions are given by:

$$\rho\left(\frac{\partial \vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u}\right) = -\nabla p$$

$$\nabla \cdot \vec{u} = 0$$

where $\vec{u}(x, y, z, t) = (u, v, w)$ is the fluid velocity, p is the pressure, and ρ is the constant density. For simplicity, we consider the equation in the form where p represents p/ρ.

We seek the most general scaling transformation group under which the Euler equations are invariant. Let the transformation be defined by a set of real exponents $\alpha_x, \alpha_t, \alpha_u, \alpha_p$ and a scaling parameter $k \in (0, \infty)$:

$$(x', y', z') = k^{\alpha_x}(x, y, z)$$
$$t' = k^{\alpha_t} t$$
$$\vec{u}' = k^{\alpha_u}\vec{u}$$
$$p' = k^{\alpha_p} p$$

We now compute how each term in equation (1) scales:

(1) The time derivative scales as:

$$\left(\frac{\partial \vec{u}}{\partial t}\right)' = \frac{\partial}{\partial t'}(k^{\alpha_u}\vec{u}) = k^{\alpha_u-\alpha_t}\frac{\partial \vec{u}}{\partial t}.$$

(2) The convective term scales as:

$$((\vec{u}\cdot\nabla)\vec{u})' = (k^{\alpha_u}\vec{u}\cdot \tfrac{1}{k^{\alpha_x}}\nabla)(k^{\alpha_u}\vec{u}) = k^{2\alpha_u-\alpha_x}(\vec{u}\cdot\nabla)\vec{u}.$$

(3) The pressure gradient term scales as:

$$\nabla p' = \tfrac{1}{k^{\alpha_x}}\nabla(k^{\alpha_p}p) = k^{\alpha_p-\alpha_x}\nabla p.$$

To preserve the form of the equation, all terms in (1) must scale identically. Therefore, we equate the exponents (the first equation means both terms in the Lhs must be multiplied by the exact same coefficient after being transformed; the second equation is a follow-up of the first to equate the exponents of Lhs and Rhs):

$$\alpha_u - \alpha_t = 2\alpha_u - \alpha_x \tag{3}$$
$$2\alpha_u - \alpha_x = \alpha_p - \alpha_x \tag{4}$$

Thus :

$$\alpha_u = \alpha_x - \alpha_t,\ \alpha_p = 2(\alpha_x - \alpha_t).$$

The most general scaling symmetry of the incompressible Euler equations is:

$$\begin{aligned}(x', y', z') &= k^{\alpha_x}(x, y, z)\\ t' &= k^{\alpha_t}t\\ \vec{u}' &= k^{\alpha_x-\alpha_t}\vec{u}\\ p' &= k^{2(\alpha_x-\alpha_t)}p\end{aligned} \tag{G.1}$$

This transformation group leaves the form of the Euler equations invariant. Notice it is identical with Eq.(2.3). In the NSE, $2\alpha_x = \alpha_t$ is the "standard scaling" - it keeps the viscosity constant and is required for many other complex reasons (see section 4). In contrast, the Euler equations possess no intrinsic physical scale (such as viscosity), and therefore admit a full two-parameter family of scaling symmetries: any pair of real exponents $(\alpha_x, \alpha_t) \in \mathbb{R}^2$ defines a valid scaling group.

We will however, borrow the NSE scaling conventions for the purpose of rightly defining large and fine scales. Under standard NSE scaling, setting $\alpha_x = 1, \alpha_t = 2$ means that the space and time correctly rescale to large scales when $k \to \infty$ and to fine scales when $k \to 0$.

G.2. **Bouton's Self-Similar Solutions for the Euler Equations.** Following Bouton's invariant theory, the requirement of invariance under the scaling group (G.1) adds two supplementary partial differential equations to the Euler system. Since the scaling exponents for $\vec{u}$ and p are identical to those in the NSE analysis, these supplementary equations are formally the same as in Eqs. 2.7, where the resulting solutions (3.2) are integral rational functions, meaning they are isobaric polynomials or ratios thereof.

G.3. **Theorem for Global Regularity of the Euler Equations.** Since we have chosen to rescale the environment variables x, y, z, t via the NSE standard scaling $\alpha_x = 1, \alpha_t = 2$, the conditions under which a self-similar solution is not subject to scaling-induced blow-up will be identical with these for the NSE. In this scenario, Lemma 6.2 also governs the regularity of Euler's equation.

We can now formulate a theorem for the global regularity of the incompressible Euler equations, mirroring Theorem 8.1 for the NSE.

Theorem G.1 (Global Regularity for Euler via Self-Similarity)**.** *Let $u_0(x, y, z)$ be any smooth, divergence-free, space-periodic initial datum or such of Schwartz-class. Then there exists a well-posed solution (u, p), which is unique, coincides with the Euler evolution of u_0 on $[0, \infty)$, remains C^∞ for all $t \geq 0$, and has all velocity, energy, and vorticity norms uniformly bounded.*

Proof.

- **Embedding of the initial data.** By the logic of Lemma 5.1 and Corollary 5.2, we can use the given smooth, divergence-free initial field $u_0(x, y, z)$ and build a one-parameter self-similar family U_0 such, that $u_0(x, y, z)$ is realized as the $k = 1$ identity scale member of U_0. When building U_0 and its corresponding initial pressure, we make sure (u, p) belong to the set (5.1) and meet the requirement of Lemma (6.2).
- **Existence and uniqueness.** Classical theorems for the Euler equations guarantee the existence and uniqueness of a smooth self-similar solution (U, P) on a maximal time interval $[0, \tau_{max})$ for smooth, diverhence-free initial data U_0 [21], [26]. The self-similar solution (U, P), being a unique solution must coincide with the Euler's evolution of U_0 on $[0, \tau_{max})$. The solution (u, p), being a unique solution must coincide with the Euler's evolution of u_0 on $[0, \tau_{max})$. (u, p) is the $k = 1$ identity scale member of (U, P).
- **A priori norm estimates.** Lemma (6.2) provides the scaling exponents which guarantee the velocity, energy and vorticity norms of (U, P) remain bounded. It shows that none of these norms grows at fine scales, which yields uniform bounds for all time. Therefore, the same applies for its $k = 1$ identity scale member (u, p).
- **Blow-up exclusion.** The classical Beale-Kato-Majda criterion for the 3D Euler equations [15] is applied to (U, P): it states that a finite-time singularity (divergence of the velocity) can occur only if the time integral of the maximum vorticity diverges. Since the a priori norm estimates ensure a uniform bound on $||\omega||$, the BKM condition $\int_0^T ||\omega(\cdot, \tau)|| d\tau < \infty$ is satisfied. This establishes the smoothness of (U, P) on $[0, \infty)$, and thus smoothness of (u, p) on $[0, \infty)$.
- **Conclusion.** Together, these steps show that the $k = 1$ self-similar member is well-posed: an unique, globally smooth solution of the incompressible Euler equations evolving from u_0, with all relevant norms remaining bounded for all $t \geq 0$. □

Here are two 3D-flow examples for the incompressible Euler equations, one space-periodic and one Schwartz-class, constructed according to the self-similarity method presented above.

G.4. **Space-Periodic Example.** Consider the 3D incompressible Euler equations:

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla)\vec{u} = -\nabla p, \qquad \nabla \cdot \vec{u} = 0.$$

G.4.1. *Bouton's Self-Similar Ansatz with T and L_i.* Following Bouton, the self-similar solution is written as:

$$\begin{aligned} \vec{u}(x, y, z, t) &= T^{\frac{\beta_x - \beta_t}{\beta_t}} \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \\ p(x, y, z, t) &= T^{\frac{2(\beta_x - \beta_t)}{\beta_t}} P\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right), \end{aligned} \tag{G.2}$$

where the exponents satisfy the NSE and Euler blow-up prevention criterion:

$$\frac{\beta_x}{\beta_t} > \frac{3}{2}.$$

G.4.2. *Choice of Exponents and Profile Function.* Choose

$$\beta_x = 5, \quad \beta_t = 3 \quad \Rightarrow \quad \frac{\beta_x}{\beta_t} = \frac{5}{3} > \frac{3}{2}.$$

Let the profile $\mathbf{F}(\xi_1, \xi_2, \xi_3, \tau)$ be smooth, divergence-free, and space-periodic in $\xi_i = x_i/L_i$:

$$\mathbf{F}(\xi_1, \xi_2, \xi_3, \tau) = U \begin{bmatrix} \sin(\xi_2) + \cos(\xi_3) \\ \sin(\xi_3) + \cos(\xi_1) \\ \sin(\xi_1) + \cos(\xi_2) \end{bmatrix} e^{-\tau}.$$

- Divergence-free: $\nabla_\xi \cdot \mathbf{F} = 0$. - Smooth and space-periodic in all directions.
—

G.4.3. *Velocity and Pressure Fields.* The self-similar velocity is

$$\vec{u}(x, y, z, t) = T^{2/3}\, \mathbf{F}\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right),$$

and the pressure

$$p(x, y, z, t) = T^{4/3}\, P\left(\frac{x}{L_1}, \frac{y}{L_2}, \frac{z}{L_3}, \frac{t}{T}\right),$$

with P determined via the incompressibility condition:

$$\nabla \cdot \vec{u} = 0 \quad \Rightarrow \quad \Delta P = -\nabla \cdot (\vec{u} \cdot \nabla \vec{u}).$$

—

G.4.4. *Vorticity and Blow-up Prevention.* The vorticity field is

$$\vec{\omega} = \nabla \times \vec{u} = T^{2/3} \nabla_\xi \times \mathbf{F}(\xi, \tau).$$

Under rescaling

$$x_i' = k^{\alpha_x} x_i = k x_i, \quad T' = k^{\alpha_t} T = k^2 T, \quad \vec{u}' = k^{2(\beta_x - \beta_t)/\beta_t} \vec{u} = k^{4/3} \vec{u},$$

the vorticity transforms as

$$\vec{\omega}' = k^{(2\beta_x - 3\beta_t)/\beta_t} \vec{\omega} = k^{1/3} \vec{\omega} \to 0 \text{ as } k \to 0,$$

ensuring no fine-scale blow-up, consistent with the BKM criterion.

G.5. **Schwartz-Class Example.** Let the initial velocity field be a smooth, rapidly decaying, divergence-free function:

$$\vec{u}_0(x, y, z) = T^{2/3} \begin{bmatrix} yze^{-r^2/L^2} \\ xze^{-r^2/L^2} \\ -2xye^{-r^2/L^2} \end{bmatrix}, \quad r^2 = x^2 + y^2 + z^2, \quad L = 1.$$

G.5.1. *Self-Similar Embedding at $k = 1$.* Embed the initial data into a self-similar family using Bouton's T-prefactor form:

$$\vec{u}(x, y, z, t) = T^{\frac{\beta_x - \beta_t}{\beta_t}} \mathbf{F}\Big(\frac{x}{L}, \frac{y}{L}, \frac{z}{L}, \frac{t}{T}\Big), \quad p(x, y, z, t) = T^{\frac{2(\beta_x - \beta_t)}{\beta_t}} F\Big(\frac{x}{L}, \frac{y}{L}, \frac{z}{L}, \frac{t}{T}\Big).$$

Choose exponents satisfying the Euler blow-up prevention criterion:

$$\beta_x = 5, \quad \beta_t = 3 \quad \Rightarrow \quad \frac{\beta_t}{\beta_x} = \frac{5}{3} > \frac{5}{2}.$$

The velocity and pressure prefactors become:

$$\vec{u} = T^{2/3} \mathbf{F}, \quad p = T^{4/3} F.$$

Explicitly, with a Gaussian profile:

$$\vec{u}(x, y, z, t) = T^{2/3}\, e^{-t/T} \begin{bmatrix} yze^{-r^2/L^2} \\ xze^{-r^2/L^2} \\ -2xye^{-r^2/L^2} \end{bmatrix}, \quad p(x, y, z, t) = T^{4/3} P(x/L, y/L, z/L, t/T).$$

At $t = 0$ the solution recovers the initial data:

$$\vec{u}(x, y, z, 0) = \vec{u}_0(x, y, z).$$

G.5.2. *Properties.*

- **Divergence-Free:** $\nabla \cdot \vec{u} = 0$
- **Time Derivative:** $\partial_t \vec{u} = -\frac{1}{T}\vec{u}$
- **Convective Term:** $(\vec{u} \cdot \nabla)\vec{u}$ is smooth and rapidly decaying
- **Pressure:** Solving $-\nabla p = \partial_t \vec{u} + (\vec{u} \cdot \nabla)\vec{u}$ yields a smooth, rapidly decaying pressure field
- **Vorticity:** $\vec{\omega} = \nabla \times \vec{u}$ scales as $\vec{\omega}' = k^{(2\beta_x - 3\beta_t)/\beta_t}\vec{\omega} = k^{1/3}\vec{\omega}$, vanishing as $k \to 0$, confirming no scaling-induced blow-up.

Author Declarations:
Conflict of Interest:
On behalf of all authors, the corresponding author states that there is no conflict of interest.
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The data that support the findings of this study are available within the article.

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